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Agilex™ 7 Power Distribution Network Design Guidelines

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1. Agilex™ 7 Power Distribution Network Design Guidelines Overview

This application note provides information for the Agilex™ 7 device family power distribution network (PDN) design guidelines. A solid design guidelines for the Agilex 7 device family PDN including fixed decoupling capacitors on board and minimum simulation is proposed.

In the previous FPGA families (for example, the Stratix® 10 and Arria® 10 devices), the PDN tool was used along with power consumption data from the Early Power Estimator (EPE) and the pin connection guidelines to design and optimize board-level PDN. However, the PDN tool is not used or supported for the later Agilex device families because it produces pessimistic results for decoupling capacitors design, particularly for the VCC core.

The information in this application note is intended to provide guidelines to allow you to successfully complete your PDN design, without requiring additional support.

Related Information

Agilex 7 Power Management User Guide
Provides more information about the power-up sequence (PUS) and power-down sequence (PDS) for the Agilex 7 devices.

1.1. Device Guidelines Status for Agilex 7 Devices

Table 1. Device Guidelines Status for Agilex 7 Devices (F-Series)

Device	Tile	Package	Guidelines Status
AGF 012/014	E-Tile & P-Tile	R24A	Final
AGF 012/014	E-Tile & P-Tile	R24B	Final
AGF 019/022/023/027	E-Tile & P-Tile	R25A	Final
AGF 006/008	F-Tile	R16A	Final
AGF 006/008/012/014/019/022/023/027	F-Tile	R24C	Final
AGF 006/008/012/014	F-Tile	R24D	Final
AGF 019/022/023/027	F-Tile	R31C	Final

Table 2. Device Guidelines Status for Agilex 7 Devices (I-Series)

Device	Tile	Package	Guidelines Status
AGI 019/023	R-Tile & F-Tile	R18A	Final
AGI 022/027	R-Tile & F-Tile	R29A	Final

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Device	Tile	Package	Guidelines Status
AGI 022/027	R-Tile & F-Tile	R31A	Final
AGI 019/022/023/027	F-Tile	R31B	Final
AGI 035/040	F-Tile	R39A	Final
AGI 041	F-Tile & R-Tile	R29D	Final
AGI 041	F-Tile	R31B	Final
AGI 041	R-Tile & F-Tile	R31E	Final

Table 3. Device Guidelines Status for Agilex 7 Devices (M-Series)

Device	Tile	Package	Status
AGM 032/039	R-Tile & F-Tile	R47A	Final
AGM 032/039	F-Tile	R31B	Final
AGM 032/039	F-Tile	R47B	Final

Guidelines status descriptions:

- Advance—for pre-production development and **subject to change**. Altera recommends the guidelines based on simulation.
- Preliminary—for pre-production development and **subject to change**. Altera recommends the guidelines based on simulation, early validation, and early characterization data.
- Final—for use in **production design**. Altera recommends the guidelines based on validation and characterization of production silicon.

2. Power Delivery Overview

This section covers the maximum power consumption budget specifically for the Agilex 7 device family. It also covers the recommended power tree or merged power rails on board to achieve minimum number of voltage regulators on board and reduce cost. The power rail names at the package-level along with their on-board (package pin) specification, rail tolerance, and the recommended step loads for the PDN time domain simulations are also covered in this section.

2.1. Power Architecture

2.1.1. Power Budget

Use the Intel® FPGA Power and Thermal Calculator (PTC) to determine the power for your applications. Scale the recommended decoupling capacitors based on the scaling factor of the exact power consumption to the maximum power consumption.

The rule of thumb is to scale the number of suggested decoupling capacitors. For example, if your scale factor is 2, and you have 3 × 47 µF capacitors as peripheral, the suggested decoupling capacitors on your board is 3/2 = (round it to 2) × 47 µF.

Ensure that the recommended voltage regulator current in the board design must be larger than the total current for the merged power rails. Add 30% margin for the voltage regulator current to ensure good reliability and thermal performance.

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2.1.2. Power Tree

This section describes the recommended power tree for the Agilex 7 device family.

There are different types of Agilex 7 devices.

- Agilex 7 (AGF) devices that cover only the E-Tile and P-Tile in the package (refer to the Recommended Agilex 7 AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices figure).
- Agilex 7 (AGF) devices that cover only the F-Tile with only FGT (general-purpose transceivers) enabled (refer to the Recommended Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and RTile) Power Tree for Production Silicon figure).
- Agilex 7 (AGI) devices that cover the R-Tile or F-Tile, or both (F-Tile has both FGT (general-purpose transceivers) enabled and FHT (high-speed transceivers) enabled (refer to the Recommended Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and RTile) Power Tree for Production Silicon figure).
- Agilex 7 (AGM) devices[1] that cover the R-Tile, F-Tile has both FGT (general-purpose transceivers) enabled and FHT (high-speed transceivers) enabled, and HBM (refer to the Recommended Agilex 7 AGM (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Engineering Sample (ES) Silicon figure).

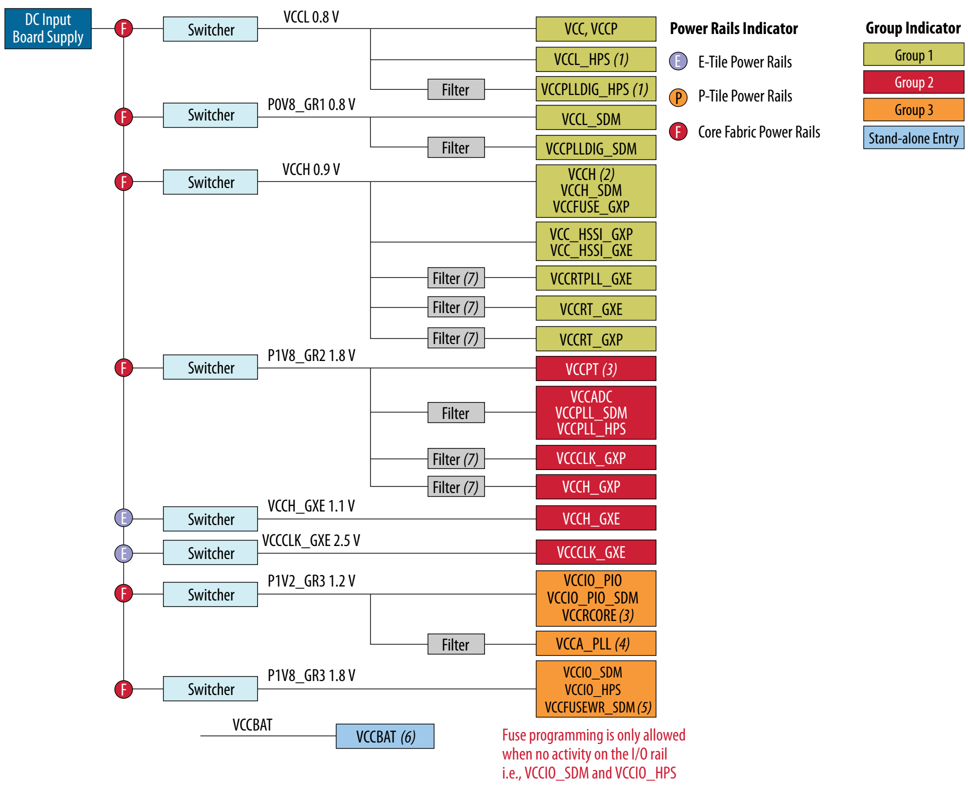
(1) The power tree information of the Agilex 7 (AGM) devices is subject to change. Intended for pre-production development, for production designs, use with caution.

2.1.2.1. Recommended Power Tree for the Agilex 7 Devices with only P-Tile and E-Tile in the Device Packages

Figure 1. Recommended Agilex 7 AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices

This power tree demonstrates the recommended FPGA PCB power rails grouping. You can use any recommended voltage regulators listed in the FPGA Core Fabric VCC Voltage Regulator Selection section as long as they meet the power rail specifications listed in the Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance table.

- This block diagram is applicable for the Agilex 7 AGF devices with E-Tile and P-Tile in the package.
- All power rails with naming _GXE refers to the E-Tile power rails and all power rails with naming _GXP refers to the P-Tile power rails.



Notes:

(1) VCCPLLDIG_HPS is always connected to VCCL_HPS via a filter in the power tree regardless of the device speed.
VCCL_HPS and VCCPLLDIG_HPS are always in Group 1 for power sequence regardless of the device speed.

- For -1V Agilex 7 device that is in the performance boost or turbo mode, VCCL_HPS and VCCPLLDIG_HPS are fed separately by an additional voltage regulator=0.95 V in Group 1.
- For -1V, -2V, -3V, and -3E Agilex 7 SmartVID devices, it is connected to VID (ranging from 0.7 V to 0.9 V). If the SmartVID is the same voltage as VCC, VCCL_HPS is connected to VCC.
- For -4F and -4X Agilex 7 devices, it is connected to a fixed voltage of 0.8 V. VCCL_HPS is merged with VCCL_SDM as both power rails are 0.8 V.

For more information, refer to the *Agilex 7 FPGAs and SoCs Device Data Sheet: F-Series and I-Series*.

(2) VCCH is 0.9 V for Agilex 7 devices with E-Tile and P-Tile in the device package (for both ES and production devices). It can be merged with VCCH_SDM on in the power tree.

(3) Refer to the VCCPT/VCCRCORE section. The VCCPT noise tolerance is +/- 5% individually but when it is merged with other power rails, the noise tolerance of the total merged rail is tighter.

(4) Refer to the VCCA_PLL section.

(5) Fuse programming is only allowed when there is no activity on the I/O rail, VCCIO_SDM, and VCCIO_HPS. Otherwise, a separate power module is required for VCCFUSEWR_SDM.

(6) Battery back-up power supply for device security Advanced Encryption Standard Battery-backed RAM (AES BBRAM) key register. When using the device security AES BBRAM key, refer to *Figure: VCCBAT Rail RC Circuit Recommendation* .
When not using the device security AES BBRAM key, it is not required to mount a non-volatile battery power source. Tie this pin to ground.

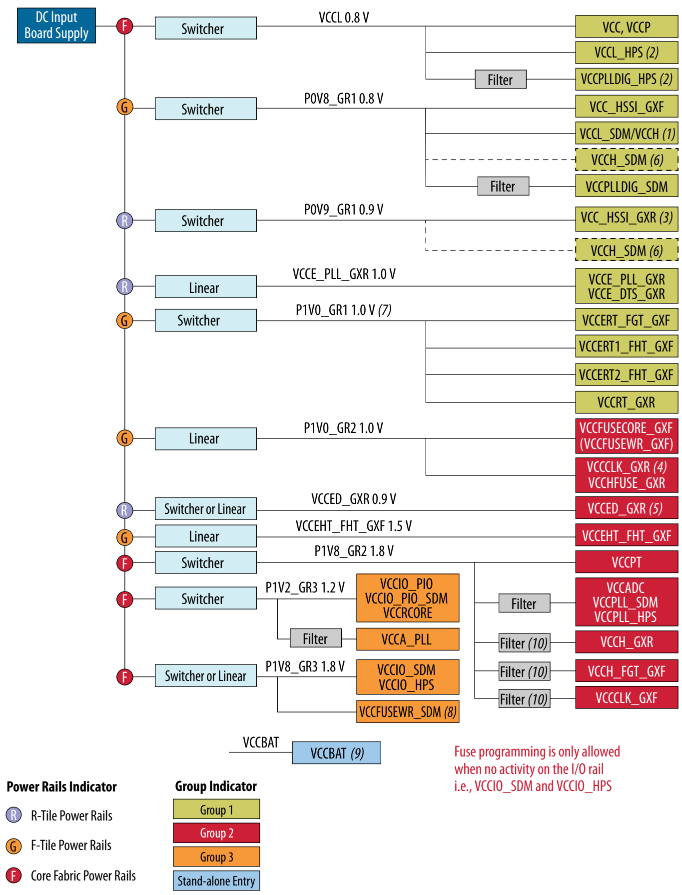
(7) Each tile must have its own LC filter.

2.1.2.2. Recommended Power Tree for the Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and R-Tile) Device Packages

Figure 2. Recommended Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon

This power tree demonstrates the recommended FPGA PCB power rails grouping. You can use any recommended voltage regulators listed in the FPGA Core Fabric VCC Voltage Regulator Selection section as long as they meet the power rail specifications listed in the Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance table.

- This block diagram is also applicable for the Agilex 7 AGF devices with only F-Tile (Only FGT transceivers enabled) in the package. However, you need to remove or disable all power rails in the power tree with power rails ending with _GXR and _FHT_GXF to exclude the R-Tile and F-Tile (FHT) from the power tree.
- All power rails with naming _GXR refers to the R-Tile power rails and all power rails with naming _GXF refers to the F-Tile power rails.



Notes:

(1) VCCH is 0.8 V for Agilex 7 production devices with F-Tile or R-Tile in device package. It can be merged with VCCH_SDM in power tree.

(2) VCCPLLDIG_HPS is always connected to VCCL_HPS via a filter in the power tree regardless of the device speed. VCCL_HPS and VCCPLLDIG_HPS are always in Group 1 for power sequence regardless of the device speed.

- For -1V Agilex 7 device that is in the performance boost or turbo mode, VCCL_HPS and VCCPLLDIG_HPS are fed separately by an additional voltage regulator=0.95 V in Group 1.
- For -1V, -2V, -3V, and -3E Agilex 7 SmartVID devices, it is connected to VID (ranging from 0.7 V to 0.9 V). If the SmartVID is the same voltage as VCC, VCCL_HPS is connected to VCC.
- For -4F and -4X Agilex 7 devices, it is connected to a fixed voltage of 0.8 V. VCCL_HPS is merged with VCCL_SDM as both power rails are 0.8 V.

For more information, refer to the *Agilex 7 FPGAs and SoCs Device Data Sheet: F-Series and I-Series*.

(3) VCC_HSSI_GXR is 0.95 V for the Agilex 7 AGIB027R29AxxxVR0/R1 device and requires a separate power rail. For other Agilex 7 devices with R-Tile, VCC_HSSI_GXR must be connected to 0.9 V and can be merged with VCCH_SDM.

(4) VCCCLK_GXR must be combined with VCCHFUSE_GXR on board. Both power rails can be merged with VCCFUSECORE_GXF.

(5) For the Agilex 7 I-Series ES and production devices, VCCED_GXR is in Group 2 and must be supplied by an independent 0.9 V voltage regulator in Group 2.

(6) VCCH_SDM is the sense power rail with very low current connecting to the VCCH circuitry at die.
For Agilex 7 devices with both F-Tile and R-Tile, you must connect this sense to the VCC_HSSI_GXR rail (0.9 V). For Agilex 7 devices with F-Tile only, you must connect this sense to the VCCH rail (0.8 V).

(7) The voltage specification for these power rails is 1 V +/-2.5% at package pin (if there is a proper voltage regulator to cover their total current with targeted voltage regulator ripple based on the required specification). Otherwise, Altera recommends you to separate these power rails supply.

(8) Fuse programming is only allowed when there is no activity on the I/O rail, VCCIO_SDM, and VCCIO_HPS. Otherwise, a separate power module is required for VCCFUSEWR_SDM.

(9) Battery back-up power supply for device security Advanced Encryption Standard Battery-backed RAM (AES BBRAM) key register. When using the device security AES BBRAM key, refer to *Figure: VCCBAT Rail RC Circuit Recommendation* .
When not using the device security AES BBRAM key, it is not required to mount a non-volatile battery power source. Tie this pin to ground.

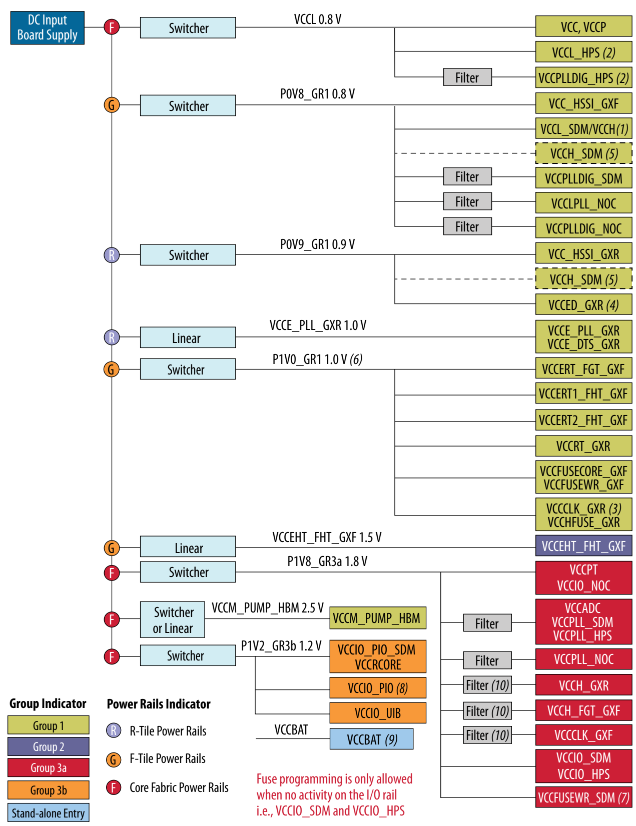
(10) Each tile must have its own LC filter.

2.1.2.3. Recommended Power Tree for the Agilex 7 AGM (with only F-Tile, or both F-Tile and R-Tile) Device Packages

Figure 3. Recommended Agilex 7 AGM (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Engineering Sample (ES) Silicon

This power tree demonstrates the recommended FPGA PCB power rails grouping. You can use any recommended voltage regulators listed in the FPGA Core Fabric VCC Voltage Regulator Selection section as long as they meet the power rail specifications listed in the Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance table.

- This block diagram is also applicable for the Agilex 7 AGM devices with only F-Tile (Only FGT transceivers enabled) in the package. However, you need to remove or disable all power rails in the power tree with power rails ending with _GXR and _FHT_GXF to exclude the R-Tile and F-Tile (FHT) from the power tree. For more information, refer to the *Unused R-Tile and F-Tile Channels* section.
- All power rails with naming _GXR refers to the R-Tile power rails and all power rails with naming _GXF refers to the F-Tile power rails.



Notes:

(1) VCCH is 0.8 V for Agilex 7 devices with F-Tile or R-Tile in the device package. It can be merged with VCCL_SDM in the power tree.

(2) VCCPLLDIG_HPS is always connected to VCCL_HPS via a filter in the power tree regardless of the device speed.
VCCL_HPS and VCCPLLDIG_HPS are always in Group 1 for power sequence regardless of device speed.

a. For -1V device and in performance boost or turbo mode, VCCL_HPS and VCCPLLDIG_HPS are fed separately by an additional voltage regulator=0.95 V in Group 1.
b. For -1V, -2V, -3V, and -3E Agilex 7 SmartVID devices, it is connected to VID (0.7 V-0.9 V).
   i. If SmartVID is the same voltage as VCC, the VCCL_HPS is connected to VCC.
c. For -4F and -4X Agilex 7 devices, it is connected to fixed voltage 0.8 V.
   i. VCCL_HPS is merged with VCCL_SDM since both are 0.8 V.

For more information, refer to the *Agilex 7 FPGAs and SoCs Device Data Sheet*.

(3) VCCCLK_GXR must be combined with VCCHFUSE_GXR on board. Both power rails can be merged with VCCFUSECORE_GXF.

(4) For the Agilex 7 M-Series ES devices, VCCED_GXR is in Group 1 and can be merged to VCC_HSSI_GXR.

(5) VCCH_SDM is the sense power rail with very low current connecting to the VCCH circuitry at die. Must connect this sense to the VCC_HSSI_GXR rail for the Agilex 7 devices with both F-Tile and R-Tile. Must connect this sense to the VCCH rail for the Agilex 7 device with F-Tile only.

(6) The voltage specification for these power rails is 1 V +/-2.5% at package pin. If there is a proper VR to cover their total current and with targeted VR ripple based on required specification. Otherwise, Altera recommends to separate these power rails supply.

(7) Fuse programming is only allowed when there is no activity on the I/O rail, VCCIO_SDM, and VCCIO_HPS. Otherwise, a separate power module is required for VCCFUSEWR_SDM.

(8) Connect these VCCIO_PIO pins to a 1.05 V, 1.1 V, 1.2 V, or 1.3 V power supplies, depending on the I/O standard required by the specific bank. If the I/O standard is not 1.2 V, separate them by an additional voltage regulator in Group 3b.

(9) Battery back-up power supply for device security Advanced Encryption Standard Battery-backed RAM (AES BBRAM) key register. When using the device security AES BBRAM key, refer to *Figure: VCCBAT Rail RC Circuit Recommendation.*

(10) Each tile must have its own LC filter.

2.2. Rail Merger Requirements

IP voltage rails of same nominal values within the same sequencing group can be merged assuming that the power delivery network to each package balls for each IP is designed with care to meet the tolerance specifications of that IP. Therefore, proper analysis or simulations should be done to ensure voltage drop and cross regulations are under control.

As many power rails are merged on the motherboard, the requirement for an LC filter is necessary to ensure systems functionality especially for rail connections to sensitive circuits such as the phase-locked loop (PLL) and clock. Altera recommends you to follow the LC filter requirements.

For more information about the rail merger requirements, refer to the *Agilex 7 Device Family Pin Connection Guidelines*.

Related Information

- Agilex 7 Device Family Pin Connection Guidelines: F-Series and I-Series
- Agilex 7 Device Family Pin Connection Guidelines: M-Series

2.3. Power Rails Specification

2.3.1. Power Nets

This section describes the Agilex 7 device family power nets and their subsystem details along with their board-level connection based on the recommended power trees described in the *Power Tree* section.

For more detailed board-level connection specification, refer to the *Agilex 7 Device Family Pin Connection Guidelines*.

Related Information

- Power Tree on page 7
- Agilex 7 Device Family Pin Connection Guidelines: F-Series and I-Series
- Agilex 7 Device Family Pin Connection Guidelines: M-Series

2.3.1.1. Agilex 7 Package Power Nets and Subsystems Details

Table 4. Agilex 7 AGF (with only E-Tile and P-Tile), Agilex 7 AGF (with only F-Tile), and Agilex 7 AGI (with only F-Tile, or both F-Tile and R-Tile) FPGA Package Power Rail Nets and Subsystem Details

Device	Power Thermal Calculator (PTC) Rail Name	Board Connections	System Connections
FPGA	VCC	VCCL 0.8 V	Fabric core
FPGA/HPS	VCCL_HPS		HPS core for –1, –2, and –3 devices. Refer to the notes in the Recommended Agilex 7 AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices and Recommended Agilex 7AGI

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Device	Power Thermal Calculator (PTC) Rail Name	Board Connections	System Connections
			or AGF (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon figures.
FPGA/HPS	VCCPLLDIG_HPS		HPS core for −1, −2, and −3 devices. Refer to the notes in the Recommended Agilex 7 AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices and Recommended Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon figures.
FPGA/PIO	VCCP		I/O 96 PHY
FPGA packages with E-Tile and P-Tile	VCCH	VCCH 0.9 V/P0V9_GR1	Rail for AIB-Bridge for FPGA packages with E-Tile and P-Tiles.
FPGA/SDM	VCCH_SDM		SDM POR monitoring ball for VCCH of FPGA packages (with E-Tile and P-Tile) or FPGA packages (with F-Tile and R-Tile).
FPGA packages (with F-Tile and R-Tile) or packages (with F-Tile only)	VCCH	P0V8_GR1	Rail for AIB-Bridge for packages with F-Tile and R-Tile or packages with F-Tile only.
FPGA/SDM	VCCH_SDM		SDM POR monitoring ball for VCCH of FPGA packages (with F-Tile only).
FPGA/SDM	VCCL_SDM		SDM core
FPGA/SDM	VCCPLLDIG_SDM		SDM digital PLL
FPGA	VCCPT	P1V8_GR2	CRAM
FPGA/SDM	VCCADC		ADC
FPGA/SDM	VCCPLL_SDM		SDM analog PLL
FPGA/HPS	VCCPLL_HPS		HPS analog PLL
FPGA/SDM	VCCIO_PIO_SDM	P1V2_GR3	SDM POR monitor for VCCIO_PIO_SDM
FPGA/PIO	VCCIO_PIO		I/O 96 I/O buffer
FPGA	VCCRCORE		Share with VCCIO_PIO only when I/O is 1.2 V.
FPGA	VCCA_PLL		Main DDR PLL
FPGA/SDM	VCCIO_SDM	P1V8_GR3	SDM 1.8 V I/O supply
FPGA/HPS	VCCIO_HPS		HPS I/O supply
FPGA/SDM	VCCBAT	VCCBAT	Battery back-up power supply for device security Advanced Encryption

continued...

Device	Power Thermal Calculator (PTC) Rail Name	Board Connections	System Connections
			Standard, Battery-backed RAM (AES BBRAM) key register.
FPGA/SDM	VCCFUSEWR_SDM	VCCFUSEWR_SDM	SDM fuse. Fuse programming is only allowed when there is no activity on the I/O rail, VCCIO_SDM, and VCCIO_HPS. Otherwise, a separate power module is required for VCCFUSEWR_SDM.

Table 5. Agilex 7 AGM (with only F-Tile, or both F-Tile and R-Tile) FPGA Package Power Rail Nets and Subsystem Details

Device	Power Thermal Calculator (PTC) Rail Name	Board Connections	System Connections
FPGA	VCC	VCCL 0.8 V	Core fabric
FPGA/HPS	VCCL_HPS		HPS core for −1, −2, and −3 devices. Refer to the notes in the Recommended Agilex 7 AGM (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Engineering Sample (ES) Silicon figure.
FPGA/HPS	VCCPLLDIG_HPS		HPS core for −1, −2, and −3 devices. Refer to the notes in the Recommended Agilex 7 AGM (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Engineering Sample (ES) Silicon figure.
FPGA/PIO	VCCP		I/O 96 PHY
FPGA/SDM	VCCH_SDM	VCC_HSSI_GXR	Must connect this sense to the VCC_HSSI_GXR rail for the Agilex 7 devices with both F-Tile and R-Tile.
FPGA packages (with F-Tile and R-Tile) or FPGA packages (with F-Tile only)	VCCH	P0V8_GR1	Rail for AIB-Bridge for packages with F-Tile and R-Tile or packages with F-Tile only.
FPGA/SDM	VCCH_SDM		SDM POR monitoring ball for VCCH of FPGA packages (with F-Tile only).
FPGA/SDM	VCCL_SDM		SDM core
FPGA/SDM	VCCPLLDIG_SDM		SDM digital PLL
FPGA/NOC	VCCLPLL_NOC		I/O and digital power pin for network-on-chip
FPGA/NOC	VCCPLLDIG_NOC		Digital power pin for network-on-chip
FPGA/UIB	VCCM_PUMP_HBM	VCCM_PUMP_HBM 2.5 V	Pump power supply for HBM2E

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Device	Power Thermal Calculator (PTC) Rail Name	Board Connections	System Connections
FPGA	VCCPT	P1V8_GR3/Group 3a	CRAM
FPGA/NOC	VCCIO_NOC		I/O power pin for network-on-chip
FPGA/SDM	VCCADC		ADC
FPGA/SDM	VCCPLL_SDM		SDM analog PLL
FPGA/HPS	VCCPLL_HPS		HPS analog PLL
FPGA/NOC	VCCPLL_NOC		Analog power pin for network-on-chip
FPGA/SDM	VCCIO_SDM		SDM 1.8 V I/O supply
FPGA/HPS	VCCIO_HPS		HPS I/O supply
FPGA/SDM	VCCFUSEWR_SDM		SDM fuse
FPGA/SDM	VCCIO_PIO_SDM	P1V2_GR3/Group 3b	SDM POR monitor for VCCIO_PIO_SDM
FPGA/PIO	VCCIO_PIO		I/O 96 I/O buffer
FPGA/UIB	VCCIO_UIB		I/O power supply for UIB
FPGA	VCCRCORE		Share with VCCIO_PIO when I/O is 1.2 V.
FPGA/SDM	VCCBAT	VCCBAT	Battery back-up power supply for device security Advanced Encryption Standard, Battery-backed RAM (AES BBRAM) key register.

Table 6. Agilex 7 E-Tile Power Rail Nets and Subsystem Details

PTC Rail Name	Board Connections	System Connections
VCCRT_GXE	VCCH 0.9 V	E-Tile TX/RX transceiver analog
VCCRTPLL_GXE		E-Tile TX/RX transceiver analog
VCC_HSSI_GXE		E-Tile TX/RX transceiver analog
VCCH_GXE	VCCH_GXE	E-Tile TX/RX transceiver analog
VCCCLK_GXE	VCCCLK_GXE	E-Tile 2.5 V I/O supply

Table 7. Agilex 7 P-Tile Power Rail Nets and Subsystem Details

PTC Rail Name	Board Connections	System Connections
VCC_HSSI_GXP	VCCH 0.9 V	P-Tile TX/RX transceiver digital
VCCFUSE_GXP		P-Tile fuse
VCCRT_GXP		P-Tile TX/RX transceiver analog
VCCH_GXP	P1V8_GR2	P-Tile TX/RX analog
VCCCLK_GXP		P-Tile 1.8 V I/O supply

Table 8. Agilex 7 R-Tile Power Rail Nets and Subsystem Details

PTC Rail Name	Board Connections	System Connections
VCC_HSSI_GXR	P0V9_GR1	• R-Tile TX/RX transceiver digital • For AGI F-Tile and R-Tile ES device, VCC_HSSI_GXR is 0.95 V, hence requires a separate power rail • For AGI F-Tile and R-Tile production device, VCC_HSSI_GXR is 0.9 V, can be merged with VCCH
VCCED_GXR	VCCED_GXR_0.9V_GR2/P0V9_GR1	R-Tile TX/RX transceiver digital. Connect to an independent 0.9 V voltage regulator power supply in Group 2 for Agilex 7 F/I-series devices with F-Tile and R-Tile. Connect to P0V9_GR1 in Group 1 for Agilex 7 M-series devices with F-Tile and R-Tile, and can be combined with VCC_HSSI_GXR.
VCCE_PLL_GXR	VCCE_PLL_GXR	R-Tile PLL reference signal power
VCCCLK_GXR	P1V0_GR2	R-Tile DFE
VCCHFUSE_GXR		R-Tile fuse sense
VCCRT_GXR	VCCRT_GXR	R-Tile TX/RX transceiver analog supply
VCCE_DTS_GXR	VCCE_PLL_GXR	—
VCCH_GXR	P1V8_GR2/P1V8_GR3	R-Tile high voltage (1.8 V). Connect to P1V8_GR2 via a filter for each Agilex 7 F/I-series devices with F-Tile and R-Tile. Connect to P1V8_GR3 via a filter for each Agilex 7 M-Series devices with F-Tile and R-Tile.

Table 9. Agilex 7 F-Tile Power Rail Nets and Subsystem Details

PTC Rail Name	Board Connections	System Connections
VCC_HSSI_GXF	P0V8_GR1	F-Tile digital power supply, connect to P0V8_GR1 without filter
VCCERT_FGT_GXF	P1V0_GR1	F-Tile general purpose analog supply
VCCERT1_FHT_GXF		F-Tile high speed (HS) analog supply
VCCERT2_FHT_GXF		F-Tile HS analog supply
VCCFUSECORE_GXF	P1V0_GR2	F-Tile fuse supply
VCCFUSEWR_GXF		F-Tile fuse sense WR
VCCEHT_FHT_GXF	VCCEHT_FHT_GXF	F-Tile HS high voltage supply
VCCH_FGT_GXF	P1V8_GR2/P1V8_GR3	F-Tile general purpose high voltage supply. Connect to P1V8_GR2 via a filter for each Agilex 7 F/I-series devices with F-Tile and R-Tile.

continued...

PTC Rail Name	Board Connections	System Connections
		Connect to P1V8_GR3 via a filter for each Agilex 7 M-Series devices with F-Tile and R-Tile.
VCCCLK_GXF		F-Tile clock buffer supply. Connect to P1V8_GR2 via a filter for each Agilex 7 F/I-series devices with F-Tile and R-Tile. Connect to P1V8_GR3 via a filter for each Agilex 7 M-Series devices with F-Tile and R-Tile.

2.3.2. Power Rails Tolerance

This section describes the power rails tolerance and budget (AC tolerance + VR accuracy) on board at package level for the Agilex 7 device family. The rail tolerance must be met at the FPGA package ball. You must consider the following instructions to measure the rail tolerance:

- VCCL (core power net) measurement is taken at the FPGA remote differential sense lines (there is assigned differential sense pins at FPGA package) with the scope set to bandwidth limited at 20 MHz.
- VCCERT1_FHT_GXF has dedicated differential sense line at package level to compensate the IR drop at die level.
- Other power rails (except for VCCL (core power)), the rail tolerance must be met at the board vias on the bottom layer directly connected to the package power balls.
- For all other power rails with the objective to compensate the IR drop on those specific power rails, place their respective voltage regulator sense point within the FPGA pin field, connecting to one of the rail's BGA pins which represents the worst IR drop on the path.

Table 10. Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance

Power Tree Rail Name	Vnom (Required) (V)	Recommended VR Accuracy (% of Vnom)	Recommended VR Ripple (% of Vnom)	Recommended AC Transient (% of Vnom)	Maximum AC Tolerance + VR Accuracy (% of Vnom)[2]
VCCL: • VCC/VCCP • VCCL_HPS • VCCPLLDIG_HPS	VID (0.68, 0.8, 0.85)	±0.5%	±2.5%		±3%
P0V8_GR1: • VCCL_SDM • VCCPLLDIG_SDM	0.8	±0.5%	±2.5%		±3%
VCCH:	0.9	±0.5%	±2.5%		±3%

continued...

(2) The specification stands for VR accuracy + (VR ripple + AC transient) rail tolerance and must be measured and met at package pin/ball.

Power Tree Rail Name	Vnom (Required) (V)	Recommended VR Accuracy (% of Vnom)	Recommended VR Ripple (% of Vnom)	Recommended AC Transient (% of Vnom)	Maximum AC Tolerance + VR Accuracy (% of Vnom)[2]
• VCCH • VCCH_SDM • VCCFUSE_GXP • VCC_HSSI_GXP • VCC_HSSI_GXE • VCCRTPLL_GXE • VCCRT_GXE • VCCRT_GXP					
P1V8_GR2[3]: • VCCPT • VCCADC • VCCPLL_SDM • VCCPLL_HPS	1.8	±0.5%	±1%[4]	±3.5%	±5%
P1V8_GR2[3]: • VCCCLK_GXP • VCCH_GXP	1.8	±0.5%	±0.5%	±2%	±3%
VCCH_GXE	1.1	±0.5%	±0.5%	±2%	±3%
VCCCLK_GXE	2.5	±0.5%	±0.5%	±3.5%	±5%
P1V2_GR3: • VCCIO_PIO • VCCIO_PIO_SDM • VCCRCORE • VCCA_PLL	1.2	±0.5%	±1%[4]	±3.5%	±5%
P1V8_GR3: • VCCIO_SDM • VCCIO_HPS • VCCFUSE_SDM	1.8	±0.5%	±1%[4]	±3.5%	±5%

Table 11. Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance

Power Tree Rail Name	Vnom (Required) (V)	Recommended VR Accuracy (% of Vnom)	Recommended VR Ripple (% of Vnom)	Recommended AC Transient (% of Vnom)	Maximum AC Tolerance + VR Accuracy (% of Vnom)[2]
VCCL:	VID (0.68, 0.8, 0.85)	±0.5%	±2.5%		±3%

continued...

(2) The specification stands for VR accuracy + (VR ripple + AC transient) rail tolerance and must be measured and met at package pin/ball.

(3) These power rails have different tolerances, when merged in one VR, use the tight tolerance requirement in board design.

(4) If you select a voltage regulator with lower than 1% voltage regulator ripple specification, the extra margin is added to the recommended AC transient specification.

Power Tree Rail Name	Vnom (Required) (V)	Recommended VR Accuracy (% of Vnom)	Recommended VR Ripple (% of Vnom)	Recommended AC Transient (% of Vnom)	Maximum AC Tolerance + VR Accuracy (% of Vnom)[2]
• VCC/VCCP • VCCL_HPS • VCCPLLDIG_HPS					
P0V8_GR1: • VCC_HSSI_GXF • VCCL_SDM • VCCPLLDIG_SDM • VCCH • VCCH_SDM	0.8	±0.5%	±2.5%		±3%
P0V9_GR1: • VCC_HSSI_GXR • VCCH_SDM	0.9	±0.5%	±2.5%		±3%
VCCE_PLL_GXR	1.0	±0.5%	±2.0%		±2.5%
P1V0_GR1[3]: • VCCERT_FGT_GXF • VCCERT1_FHT_GXF • VCCERT2_FHT_GXF • VCCRT_GXR [5][6]	1.0	±0.5%	±2.0%		±2.5%
P1V0_GR2[3]: • VCCFUSECORE_GXF/ VCCFUSEWR_GXF • VCCCLK_GXR	1.0	±0.5%	±2.5%		±3%
P1V0_GR2[3]: • VCCHFUSE_GXR	1.0	±0.5%	±1%[4]	±3.5%	±5%
VCCED_GXR	0.9	±0.5%	±2.5%		±3%
VCCEHT_FHT_GXF	1.5	±0.5%	±2.0%		±2.5%
P1V8_GR2[3]: • VCCPT • VCCADC • VCCPLL_SDM • VCCPLL_HPS	1.8	±0.5%	±1%[4]	±3.5%	±5%
P1V8_GR2[3]: • VCCH_GXR • VCCH_FGT_GXF [7]	1.8	±0.5%	±2.0%		±2.5%
P1V8_GR2[3]:	1.8	±0.5%	±2.5%		±3%

continued...

(5) The P1V0_GR1 specification for these individual specific power rails are ±2.5% (AC tolerance + VR accuracy) at package pin.This specification is also up to 1 MHz. The voltage regulator ripple specification is 5 mV p-p maximum. The AC noise specification above 1 MHz at package pin is ±15 mV (30 mV p-p). This can be measured by a high pass filter probe above 1 MHz.

(6) The VCCERT_FGT_GXF, VCCERT1_FHT_GXF, VCCERT2_FHT_GXF, and VCCRT_GXR can be merged to P1V0_GR1. The final specification of the merged power rail is the most stringent among those power rails.

Power Tree Rail Name	Vnom (Required) (V)	Recommended VR Accuracy (% of Vnom)	Recommended VR Ripple (% of Vnom)	Recommended AC Transient (% of Vnom)	Maximum AC Tolerance + VR Accuracy (% of Vnom)[2]
• VCCCLK_GXF					
P1V2_GR3: • VCCIO_PIO • VCCIO_PIO_SDM • VCCRCORE • VCCA_PLL	1.2	±0.5%	±1%[4]	±3.5%	±5%
P1V8_GR3: • VCCIO_SDM • VCCIO_HPS • VCCFUSEWR_SDM	1.8	±0.5%	±1%[4]	±3.5%	±5%

Table 12. Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance

Power Tree Rail Name	Vnom (Required) (V)	Recommended VR Accuracy (% of Vnom)	Recommended VR Ripple (% of Vnom)	Recommended AC Transient (% of Vnom)	Maximum AC Tolerance + VR Accuracy (% of Vnom)[2]
VCCL: • VCC/VCCP • VCCL_HPS • VCCPLLDIG_HPS	VID (0.68, 0.8, 0.85)	±0.5%	±2.5%		±3%
P0V8_GR1: • VCC_HSSI_GXF • VCCL_SDM • VCCPLLDIG_SDM • VCCH • VCCH_SDM • VCCLPLL_NOC • VCCPLLDIG_NOC	0.8	±0.5%	±2.5%		±3%
P0V9_GR1: • VCC_HSSI_GXR • VCCH_SDM • VCCED_GXR	0.9	±0.5%	±2.5%		±3%
VCCE_PLL_GXR	1.0	±0.5%	±2.0%		±2.5%
P1V0_GR1[3]: • VCCERT_FGT_GXF • VCCERT1_FHT_GXF • VCCERT2_FHT_GXF • VCCRT_GXR	1.0	±0.5%	±2.0%		±2.5%

continued...

(7) The VCCH_FGT_GXF specification for this individual power rail per *Agilex 7 FPGAs and SoCs Device Data Sheet: F-Series and I-Series* is ±2.5% (VR accuracy + AC tolerance). This applies up to 1 MHz (use of probe with low pass filter at 1 MHz at package pin, the voltage regulator ripple specification is 5 mV p-p maximum). The AC noise specification above 1 MHz at package pin is ±15 mV (30 mV p-p). This can be measured by a high pass filter probe above 1 MHz.

Power Tree Rail Name	Vnom (Required) (V)	Recommended VR Accuracy (% of Vnom)	Recommended VR Ripple (% of Vnom)	Recommended AC Transient (% of Vnom)	Maximum AC Tolerance + VR Accuracy (% of Vnom)(2)
(5)(6)					
P1V0_GR1(3): • VCCFUSECORE_GXF (7) / VCCFUSEWR_GXF • VCCCLK_GXR	1.0	±0.5%	±2.5%		±3%
P1V0_GR1(3): • VCCHFUSE_GXR	1.0	±0.5%	±1%(4)	±3.5%	±5%
VCCEHT_FHT_GXF	1.5	±0.5%	±2.0%		±2.5%
P1V8_GR3a(3): • VCCPT • VCCADC • VCCPLL_SDM • VCCPLL_HPS • VCCIO_SDM • VCCIO_HPS • VCCFUSEWR_SDM	1.8	±0.5%	±1%(4)	±3.5%	±5%
P1V8_GR3a(3): • VCCH_GXR • VCCH_FGT_GXF(7)	1.8	±0.5%	±2.0%		±2.5%
P1V8_GR3a(3): • VCCCLK_GXF	1.8	±0.5%	±2.5%		±3%
P1V8_GR3a(3): • VCCPLL_NOC • VCCIO_NOC	1.8	±0.5%	±0.5%	±2.0%	±3%
VCCM_PUMP_HBM	2.5	±0.5%	±1%(4)	±3.5%	±5%
P1V2_GR3b(3): • VCCIO_PIO • VCCIO_PIO_SDM	1.2	±0.5%	±1%(4)	±1.5%	±3%
P1V2_GR3b(3): • VCCRCORE • VCCIO_UIB	1.2	±0.5%	±1%(4)	±3.5%	±5%

The actual specification for each power rail at package level is listed in the *Agilex 7 FPGAs and SoCs Device Data Sheet: F-Series and I-Series*. To reduce the PCB cost, some power rails are merged by using only a single voltage regulator to feed the power rails. The AC + DC specification in this design guideline is based on this design. However, you have the option to not combine the power rails on the PCB in your design.

You can combine the analog power rails on the PCB (provided that the nominal voltages are the same) by using a single voltage regulator on the PCB to feed the power rails to achieve the minimum cost. If you implement this design, you must ensure the power rail specifications of the combined power rails on the PCB and voltage regulator must follow the most stringent specification of those package power rails.

The analog power rails can also be merged with the digital power rails if they are the same voltage and are in the same power-up or power-down sequence. However, ensure using the noise isolation filter from the digital power rails to analog power rails (since they are more sensitive to noise).

The Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables shows the power rail tolerance (AC tolerance + VR accuracy) based on the power tree in the Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance figure.

If a different power tree in the Recommended Power Tree for the Devices with only P-Tile and E-Tile in the Device Packages figure is used, the rail tolerance of each power net must fall into their recommended grouping category in the Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables.

2.3.3. Power Nets and Transient Specifications

Rail transient provided in the *Agilex 7 Device Family Transient and Step Load Specifications at Package Pin* table is used to design and simulate the board level. Choose the recommended load slew rates and step load at FPGA package ball below for PCB-level PDN system simulations and design. The Agilex 7 Device Family Transient and Step Load Specifications at Package Pin table shows the maximum tolerable step load at FPGA package pin. The recommended step load in the Agilex 7 Device Family Transient and Step Load Specifications at Package Pin table is connected to FPGA package ball along with the PCB post-layout model (with decoupling capacitors and voltage regulator model excluding package and silicon/die model) in an EDA tool for time domain simulation to meet rail tolerance of respective power net in the Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables at FPGA package ball.

The Agilex 7 Device Family Transient and Step Load Specifications at Package Pin table shows for the recommended step load at package ball and step load's slew rate.

Note: The step load suggested in the Agilex 7 Device Family Transient and Step Load Specifications at Package Pin table is for a normal application and worst case operation in FPGA and tiles. Step load can also be scaled if the dynamic current of the target power rail in PTC is smaller than the step load in the Agilex 7 Device Family Transient and Step Load Specifications at Package Pin table.

Table 13. Agilex 7 Device Family Transient and Step Load Specifications at Package Pin

Package Power Rails	At Package Balls (Step Load)	DI/dt at Package Balls (for Board Design)-Slew Rate	Notes
	DI (A)-Step Load	DI/dt (A/µs)-Slew Rate	
VCC/VCCP Core: AGF006/AGF008	4	20	The step load is the worst step load in the design based on 80% core utilization, 80% DSP, and 30% M20K utilization. The toggling rate is assumed to be 15%.
VCC/VCCP Core: AGF012/AGF014	17	200	The step load is the worst step load in the design based on 80% core utilization, 80% DSP, and 30% M20K utilization. The toggling rate is assumed to be 15%.
VCC/VCCP Core: AGF012/AGF014 (low power scenario)	12	141	Low power scenario case, maximum current at 70 A.
VCC/VCCP Core: AGF019/AGF023 and AGI019/AGI023	30.5	305	The step load is the worst step load in the design based on 80% core utilization, 80% DSP, and 30% M20K utilization. The toggling rate is assumed to be 15%.
VCC/VCCP Core: AGF027/AGF022 and AGI027/AGI022	32.5	325	The step load is the worst step load in the design based on 80% core utilization, 80% DSP, and 30% M20K utilization. The toggling rate is assumed to be 15%.
VCC/VCCP Core: AGI035/AGI040	7	269	The step load is the worst step load in the design based on 80% core utilization, 80% DSP, and 30% M20K utilization. The toggling rate is assumed to be 15%.
VCC/VCCP Core: AGI041	21	420	The step load is the worst step load in the design based on 80% core utilization, 80% DSP, and 30% M20K utilization. The toggling rate is assumed to be 15%.
VCC/VCCP Core: AGM039/AGM032 (R47A)	23	742	The step load is the worst step load in the design based on 80% core utilization, 80% DSP, and 30% M20K utilization. The toggling rate is assumed to be 15%.
VCC/VCCP Core: AGM039 (R31B)	10	250	The step load is the worst step load in the design based on 80% core utilization, 80% DSP, and 30% M20K utilization. The toggling rate is assumed to be 15%.
VCCPT	2.4	12	—
VCCPT (low power scenario)	0.22	1.1	Low power scenario case, maximum current at 0.8 A.

continued...

Package Power Rails	At Package Balls (Step Load)	DI/dt at Package Balls (for Board Design)-Slew Rate	Notes
	DI (A)-Step Load	DI/dt (A/µs)-Slew Rate	
VCCIO_PIO	0.645	10.8	Current specification is per I/O bank. Each I/O bank consists of 96 x I/Os. More I/O banks can join the same voltage regulator but current specification stays per I/O bank.
VCCH	1.12	4.8	Step current at package ball per each AIB Bridge. The number of AIB Bridge is calculated based on number of tiles in package.
VCCRT_GXE	0.88	35.2	Per E-Tile
VCCRT_GXE (low power scenario)	0.44	5.86	Per 2 quads of E-Tile, low power scenario case, maximum current at 3 A.
VCCRTPLL_GXE	0.3	6	Per E-Tile
VCC_HSSI_GXP	1.6	20	Per P-Tile
VCC_HSSI_GXP (low power scenario)	1.0	10	Per P-Tile, low power scenario case, maximum current at 3.5 A.
VCCRT_GXP	1.56	14.85	Slowest step load but largest current amplitude. Per P-Tile.
VCCRT_GXP (low power scenario)	0.3	6	Per 2 quads of P-Tile, low power scenario case, maximum current at 1.5 A.
VCCH_GXP	0.37	50	Per P-Tile
VCCH_GXP (low power scenario)	0.2	50	Per P-Tile, low power scenario case, maximum current at 0.7 A.
VCC_HSSI_GXF	0.825	30.55	Per F-Tile
VCCERT_FGT_GXF	0.228	22.8	Per single FGT—F-Tile channel
	0.935	1.33	For 8 x FGT—F-Tile channels
	1.87	1.24	For 16 x FGT—F-Tile channels
VCCH_FGT_GXF	0.031	31	Per single FGT—F-Tile channel
	0.18	0.26	For 8 x FGT—F-Tile channels
	0.37	0.25	For 16 x FGT—F-Tile channels
VCCERT2_FHT_GXF	0.03	0.49	Per single FHT—F-Tile channel
	0.128	0.43	For 4 x FHT—F-Tile channels
VCCERT1_FHT_GXF	0.207	3.04	Per single FHT—F-Tile channel
	0.785	2.62	For 4 x FHT—F-Tile channels
VCCEHT_FHT_GXF	0.022	0.35	Per single FHT—F-Tile channel
	0.064	0.21	For 4 x FHT—F-Tile channels

continued...

Package Power Rails	At Package Balls (Step Load)	DI/dt at Package Balls (for Board Design)-Slew Rate	Notes
	DI (A)-Step Load	DI/dt (A/μs)-Slew Rate	
VCC_HSSI_GXR	1.585	52.83	Per R-Tile
VCCRT_GXR	2.47	6.58	
VCCH_GXR	0.026	12.56	

You must also notice the following:

1. Step current at package pin is only provided for critical power rails due to either having high current/power profile at die or being highly sensitive. Altera recommends you to do transient/time domain PDN simulation by using this step load for these critical power rails in the Agilex 7 Device Family Transient and Step Load Specifications at Package Pin table to ensure you can meet the voltage specification at package pin. If the voltage specification is not met at package pin, decoupling capacitors must be adjusted.
2. Altera do not provide step current for other power rails not mentioned in the Agilex 7 Device Family Transient and Step Load Specifications at Package Pin table. Those power rails are called non-critical power rail due to having less sensitivity or low current profile/power consumption at silicon. Altera do not recommend time domain PDN analysis for non-critical power rails. Non-critical power rails PDN design suggested in this application note is guaranteed.
3. Altera recommends you to perform DC IR drop analysis for all power rails.

2.4. Decoupling Capacitors Recommendation

The recommended decoupling capacitors for FPGA on board-level PCBs are listed in this section in table format for all power nets, based on the maximum power consumption of the FPGA and suggested power tree configurations. You must select the voltage regulator capacitors (bulk decoupling capacitors) based on the data sheet specification from the voltage regulator supplier and required specifications by Altera such as the maximum voltage regulator ripple and maximum total current support for that specific power rail or combined power rails. You must perform the PDN transient simulation if your design is not following the recommendation.

Follow the recommended decoupling capacitors, power rail grouping, and recommended voltage regulator on the PCB to meet power rail tolerance and specifications at the package ball. For boards that use less power than the maximum, scale the decoupling capacitors by the radio of board's power consumption to the maximum power per power rail. However, you must perform transient PDN simulations to verify power rail tolerance at the package ball.

2.4.1. Agilex 7 FPGA Packages Board-Level Decoupling Capacitors Summary

The Agilex 7 FPGA F-Series, I-Series, and M-Series Decoupling Capacitors Summary table shows the PCB recommended FPGA decoupling capacitor requirement for the Agilex 7 AGF006/AGF008 (small package size), Agilex 7 AGF012/AGF014 (medium package size), Agilex 7 AGI041/AGI040/AGI035/AGI027/AGI023/AGI022/AGI019, Agilex 7 AGF027/AGF023/AGF022/AGF019, and Agilex 7 AGM039/AGM032 large device packages.

Fabric or core decoupling capacitors are highly dependent on the size of the core or fabric (LE and number of GPIO pins) in the package.

Decoupling capacitor recommendations for each tile is valid for any size of packages.

Table 14. Agilex 7 FPGA F-Series, I-Series, and M-Series Decoupling Capacitors Summary

Device	Agilex 7 Power and Thermal Calculator (PTC) Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
		Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	
FPGA AGF006/008 (R16A, R24C and R24D) Fabric	Merged VCC and VCCP	3x 47uF 0805	3x 47uF 0805	3x 47uF 0805	3x 47uF 0805	VR vendors capacitors recommendation must be compiled. Place 47uF 0805 capacitors inside bottom-side cavity. Use 47uF 0805 periphery capacitors on top-layer near FPGA.
FPGA AGF012/014 (R24A, R24B, R24C and R24D) Fabric	Merged VCC and VCCP	9x 47uF 0805 30x 10uF 0402 DNI(8)	5x 47uF 0805 30x 10uF 0402 DNI(8)	6x 47uF 0805	3x 47uF 0805	VR vendors capacitors recommendation must be compiled. Place thick 0805 (47uF) capacitors inside bottom-side cavity. Use 47uF 0805 periphery capacitors on top-layer near FPGA. Place 0402 capacitors in the FPGA pin field.

continued...

(8) DNI: Do not install these capacitors but include the footprints of these capacitors. This is for defensive design for additional decoupling capacitance in the event the cavity capacitor (0805) is not sufficient.

Device	Agilex 7 Power and Thermal Calculator (PTC) Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
		Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	
FPGA AGF012/014 (R24A, R24B, R24C and R24D) Fabric	Merged VCC and VCCP (low power scenario)	N/A	2x 100uF 0805 4x 47uF 0805 20x 10uF 0402 DNI(8)	N/A	N/A	For the low power scenario case, the maximum current is at 70 A. Thin board only for PCIe* FF. You must comply with the capacitors recommendation provided by the voltage regulators vendors. Place thick 0805 (47uF and 100uF) capacitors inside the bottom-side cavity. Place 0402 capacitors in the FPGA pin field.
FPGA AGF019/ AGF023 Fabric (R25A, R24C, R31C) and AGI019/ AGI023 (R31B, R18A) Fabric	Merged VCC and VCCP	6x 100uF 0805 6x 47uF 0805 30x 10uF 0402 DNI(8)	4x 100uF 0805 4x 47uF 0805 20x 10uF 0402 DNI(8)	6x 100uF 0805 6x 47uF 0805	4x 100uF 0805 4x 47uF 0805	VR vendors capacitors recommendation must be compiled. Place thick 0805 (47uF and 100uF) capacitors inside bottom-side cavity. Use 47uF and 100uF 0805 periphery capacitors on top-layer near FPGA. Place 0402 capacitors in the FPGA pin field.
FPGA AGF022/ AGF027 Fabric (R25A, R24C, R31C) and AGI022/ AGI027 (R29A, R31A, R31B)Fabric	Merged VCC and VCCP	8x 100uF 0805 4x 47uF 0805 40x 10uF 0402 DNI(8)	6x 100uF 0805 4x 47uF 0805 30x 10uF 0402 DNI(8)	12x 100uF 0805 10x 47uF 0805	10x 100uF 0805 8x 47uF 0805	VR vendors capacitors recommendation must be compiled. Place thick 0805 (47uF and 100uF) capacitors inside bottom-side cavity.

continued...

Device	Agilex 7 Power and Thermal Calculator (PTC) Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
		Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	
						Use 47uF and 100uF 0805 periphery capacitors on top-layer near FPGA. Place 0402 capacitors in the FPGA pin field.
FPGA AGI035/ AGI040 (R39A) Fabric	Merged VCC and VCCP	14x 100uF 0805 50x 10uF 0402 DNI(8)	10x 100uF 0805 50x 10uF 0402 DNI(8)	24x 100uF 0805	20x 100uF 0805	VR vendors capacitors recommendation must be compiled. Place thick 0805 (100uF) capacitors inside bottom-side cavity. Use 100uF 0805 periphery capacitors on top-layer near FPGA. Place 0402 capacitors in the FPGA pin field.
FPGA AGI041 (R29D, R31B, R31E) Fabric	Merged VCC and VCCP	19x 100uF 0805 50x 10uF 0402 DNI(8)	15x 100uF 0805 30x 10uF 0402 DNI(8)	17x 100uF 0805	13x 100uF 0805	VR vendors capacitors recommendation must be compiled. Place thick 0805 (100uF) capacitors inside bottom-side cavity. Use 100uF 0805 periphery capacitors on top-layer near FPGA. Place 0402 capacitors in the FPGA pin field.
FPGA AGM039/ AGM032 (R47A) Fabric	Merged VCC and VCCP	18x 100uF 0805 50x 10uF 0402 DNI(8)	15x 100uF 0805 30x 10uF 0402 DNI(8)	55x 100uF 0805	50x 100uF 0805	VR vendors capacitors recommendation must be compiled.

continued...

Device	Agilex 7 Power and Thermal Calculator (PTC) Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
		Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	
						Place thick 0805 (100uF) capacitors inside bottom-side cavity. Use 100uF 0805 periphery capacitors on top-layer near FPGA. Place 0402 capacitors in the FPGA pin field.
FPGA AGM039 (R31B) Fabric	Merged VCC and VCCP	10x 100uF 0805 30x 10uF 0402 DNI[8]	10x 100uF 0805 30x 10uF 0402 DNI[8]	40x 100uF 0805	40x 100uF 0805	VR vendors capacitors recommendation must be compiled. Place thick 0805 (100uF) capacitors inside bottom-side cavity. Use 100uF 0805 periphery capacitors on top-layer near FPGA. Place 0402 capacitors in the FPGA pin field.
FPGA AGI, AGF, and AGM/HPS	VCCPLLDIG_HPS	1x 1uF 0201	1x 1uF 0201	LC filter capacitors	LC filter capacitors	—
FPGA AGI, AGF, and AGM/HPS	VCCL_HPS	2x 10uF 0402 or 3x 4.7uF 0201	2x 10uF 0402 or 3x 4.7uF 0201	1x 22uF 0603	1x 22uF 0603	For SoC-centric designs.
FPGA AGI, AGF, and AGM	VCCH	4x 22uF 0603	4x 22uF 0603	2x 47uF 0805	2x 47uF 0805	Place thick 0603 (22uF) capacitors inside bottom-side cavity.
FPGA AGI, AGF, and AGM/SDM	VCCH_SDM	1x 1uF 0201	1x 1uF 0201	N/A	N/A	—
FPGA AGI, AGF, and AGM/SDM	VCCL_SDM	2x 1uF 0201	2x 1uF 0201	N/A	N/A	—
FPGA AGI, AGF, and AGM/SDM	VCCPLLDIG_SDM	1x 1uF 0201	1x 1uF 0201	LC filter capacitors	LC filter capacitors	—
FPGA AGI, AGF, and AGM	VCCPT	2x 4.7uF 0201	2x 4.7uF 0201	N/A	N/A	
FPGA AGI, AGF, and AGM/PIO	VCCRCORE	1x 1uF 0201	1x 1uF 0201	N/A	N/A	—

continued...

Device	Agilex 7 Power and Thermal Calculator (PTC) Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
		Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	
FPGA AGI, AGF, and AGM/PIO	VCCA_PLL	2x 10uF 0402	2x 10uF 0402	LC filter capacitors	LC filter capacitors	This is applicable to both reasonable worst case and low power scenario case. For the low power scenario case, the maximum current is 1 A, and the difference is the LC filter capacitors. For more information, refer to the VCCA_PLL Filter Recommendation and VCCA_PLL LC Filter Recommendation (Low Power Scenario) figures.
FPGA AGI, AGF, and AGM/SDM	VCCADC	1x 1uF 0201	1x 1uF 0201	LC filter capacitors	LC filter capacitors	—
FPGA AGI, AGF, and AGM/SDM	VCCPLL_SDM	1x 1uF 0201	1x 1uF 0201	LC filter capacitors	LC filter capacitors	—
FPGA AGI, AGF, and AGM/HPS	VCCPLL_HPS	1x 1uF 0201	1x 1uF 0201	LC filter capacitors	LC filter capacitors	—
FPGA AGI, AGF, and AGM/SDM	VCCIO_PIO_SDM	1x 1uF 0201	1x 1uF 0201	N/A	N/A	—
FPGA AGI, AGF, and AGM/PIO	VCCIO_PIO	2x 4.7uF 0402	2x 4.7uF 0402	1x 10uF 0402	1x 10uF 0402	Per each I/O bank
FPGA AGI, AGF, and AGM/SDM	VCCIO_SDM	1x 1uF 0201	1x 1uF 0201	1x 10uF 0402	1x 10uF 0402	—
FPGA AGI, AGF, and AGM/HPS	VCCIO_HPS	1x 1uF 0201	1x 1uF 0201	N/A	N/A	—
FPGA AGI, AGF, and AGM/SDM	VCCBAT	1x 1uF 0201	1x 1uF 0201	1x 10uF 0402	1x 10uF 0402	—
FPGA AGI, AGF, and AGM/SDM	VCCFUSEWR_SDM	1x 1uF 0201	1x 1uF 0201	1x 4,7uF 0201	1x 4,7uF 0201	—
FPGA AGM/UIB	VCCIO_UIB	1x 10uF 0402	1x 10uF 0402	2x 47uF 0805	2x 47uF 0805	Per HBM
FPGA AGM/UIB	VCCM_PUMP_HBM	1x 10uF 0402	1x 10uF 0402	N/A	N/A	Per HBM

continued...

Device	Agilex 7 Power and Thermal Calculator (PTC) Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
		Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (>65 mil thickness)	Thin PCB (≤65 mil thickness)	
FPGA AGM/NOC	VCCIO_NOC	1x 1uF 0402	1x 1uF 0402	N/A	N/A	Per side
FPGA AGM/NOC	VCCPLL_NOC	1x 2.2uF 0402	1x 2.2uF 0402	LC filter capacitors	LC filter capacitors	Per side
FPGA AGM/NOC	VCCLPLL_NOC	1x 2.2uF 0402	1x 2.2uF 0402	LC filter capacitors	LC filter capacitors	Per side
FPGA AGM/NOC	VCCPLLDIG_NOC	1x 2.2uF 0402	1x 2.2uF 0402	LC filter capacitors	LC filter capacitors	Per side

2.4.2. Agilex 7 E-Tile Decoupling Capacitors Summary per TileAgilex 7 E-Tile Board-Level Decoupling Capacitors Summary

Table 15. Agilex 7 E-Tile Decoupling Capacitors Summary per Tile

Agilex 7 PTC Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
	Thick PCB (≥65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (≥65 mil thickness)	Thin PCB (≤65 mil thickness)	
VCCRT_GXE	6x 4.7uF 0201	6x 4.7uF 0201	LC filter capacitors	LC filter capacitors	Per each E-Tile. This is applicable to both reasonable worst case and low power scenario case. For the low power scenario case, the maximum current is 3 A, and the difference is the LC filter capacitors. For more information, refer to the Connection Requirement and Filter Recommendation for VCCRT_GXE per Tile and Connection Requirement and Filter Recommendation for VCCRT_GXE per Tile (Low Power Scenario) figures.
VCCRTPLL_GXE	2x 1uF 0201	2x 1uF 0201	LC filter capacitors	LC filter capacitors	Per each E-Tile
VCC_HSSI_GXE	3x 10uF 0402 or 10x 4.7uF 0201	3x 10uF 0402 or 10x 4.7uF 0201	N/A	N/A	Per each E-Tile
VCCH_GXE	2x 4.7uF 0201	2x 4.7uF 0201	1x 22uF 0603	1x 22uF 0603	Per each E-Tile
VCCCLK_GXE	1x 1uF 0201	1x 1uF 0201	N/A	N/A	Per each E-Tile

Although it is not required to use the recommended decoupling capacitors and LC filters for unused E-Tile, Altera recommends you to use at least one of each suggested decoupling capacitors for the bottom-side capacitors if the E-Tile is not used.

If you plan to run the E-Tile based on OTN protocol standards, contact Altera Support and quote ID #1509475736 to receive a step-by-step guideline about updating or adjusting the PCB decoupling capacitors for E-Tile (only for VCCRT_GXE and VCCRTPLL_GXE power rails) to support the OTN application.

2.4.3. Agilex 7 P-Tile Board-Level Decoupling Capacitors Summary

Table 16. Agilex 7 P-Tile Decoupling Capacitors Summary per Tile

Agilex 7 PTC Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
	Thick PCB (≥65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (≥65 mil thickness)	Thin PCB (≤65 mil thickness)	
VCC_HSSI_GXP	1x 10uF 0402	1x 10uF 0402	N/A	N/A	Per each P-Tile
VCCFUSE_GXP	1x 1uF 0201	1x 1uF 0201	N/A	N/A	Per each P-Tile. For the multiple P-Tiles in the device package, use 1x 0402 4.7uF per 2 P-Tiles.
VCCRT_GXP	6x 4.7uF 0201	6x 4.7uF 0201	LC filter capacitors	LC filter capacitors	Per each P-Tile. This is applicable to both reasonable worst case and low power scenario case. For the low power scenario case, the maximum current is 1.5 A, and the difference is the LC filter capacitors. For more information, refer to the Filter Recommendation for VCCRT_GXP per Tile and Filter Recommendation for VCCRT_GXP per Tile (Low Power Scenario) figures.
VCCH_GXP	1x 1uF 0201	1x 1uF 0201	LC filter capacitors	LC filter capacitors	Per each P-Tile. This is applicable to both reasonable worst case and low power scenario case. For the low power scenario case, the maximum current is 0.7 A, and the

continued...

Agilex 7 PTC Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
	Thick PCB (≥65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (≥65 mil thickness)	Thin PCB (≤65 mil thickness)	
					difference is the LC filter capacitors. For more information, refer to the Filter Recommendation for VCCH_GXP per Tile and Filter Recommendation for VCCH_GXP per Tile (Low Power Scenario) figures.
VCCCLK_GXP	1x 1uF 0201	1x 1uF 0201	LC filter capacitors	LC filter capacitors	Per each P-Tile

Although it is not required to use the recommended decoupling capacitors and LC filters for unused P-Tile, Altera recommends you to use at least one of each suggested decoupling capacitors for the bottom-side capacitors if the P-Tile is not used.

2.4.4. Agilex 7 F-Tile Board-Level Decoupling Capacitors Summary

Table 17. Agilex 7 F-Tile Decoupling Capacitors Summary per Tile

Agilex 7 PTC Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
	Thick PCB (≥65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (≥65 mil thickness)	Thin PCB (≤65 mil thickness)	
VCC_HSSI_GXF	2x 10uF 0402		2x 47uF 0805		Per each F-Tile
VCCERT_FGT_GXF	2x 10uF 0402		4x 47uF 0805		Per each F-Tile
VCCERT2_FHT_GXF	1x 10uF 0402		N/A		Per each F-Tile
VCCERT1_FHT_GXF	2x 10uF 0402		2x 47uF 0805		Per each F-Tile
VCCH_FGT_GXF	2x 10uF 0402		N/A		Per each F-Tile
VCCEHT_FHT_GXF	2x 10uF 0402		N/A		Per each F-Tile
VCCCLK_GXF	2x 10uF 0402		N/A		Per each F-Tile
VCCFUSEWR_GXF	1x 1uF 0201		N/A		Per each F-Tile
VCCFUSECORE_GXF	1x 4.7uF 0201		N/A		Per each F-Tile

2.4.5. Agilex 7 R-Tile Board-Level Decoupling Capacitors Summary

Table 18. Agilex 7 R-Tile Decoupling Capacitors Summary per Tile

Agilex 7 PTC Rail Name	Bottom-side Capacitors		FPGA Periphery Capacitors		Notes
	Thick PCB (≥65 mil thickness)	Thin PCB (≤65 mil thickness)	Thick PCB (≥65 mil thickness)	Thin PCB (≤65 mil thickness)	
VCC_HSSI_GXR	2x 10uF 0402		2x 47uF 0805		Per each R-Tile
VCCE_PLL_GXR	1x 4.7uF 0201		N/A		Per each R-Tile
VCCE_DTS_GXR					
VCCRT_GXR	4x 10uF 0402		2x 47uF 0805		Per each R-Tile.
	4x 10uF 0402		2x 220uF 1206 + 3x 100uF 0805		Per each R-Tile: If LDO voltage regulator is used for this power rail.
VCCH_GXR	1x 10uF 0402		1x 47uF 0805		Per each R-Tile.
VCCED_GXR	1x 10uF 0402		1x 47uF 0805		Per each R-Tile
VCCCLK_GXR	2x 10uF 0402		N/A		Per each R-Tile
VCCHFUSE_GXR	2x 10uF 0402		N/A		Per each R-Tile

The capacitors listed in the tables above have been validated on the Altera development kits and should be used as a reference only. You have the option to choose other capacitors according to specific applications, package size used, temperature range, form factor, and rated voltage, as long as the specifications stated above are met and the LC filter works well. Altera recommends you to run the simulation to ensure there is no further changes.

This recommended decoupling capacitors solution has been verified along with the use of the recommended switching voltage regulator or LDO in the example power trees. You can use other switching voltage regulators or LDOs as long as you meet the PCB power rail tolerance specification in the Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables. Altera recommends performing the PCB time domain simulation by using the step load in the Agilex 7 Device Family Transient and Step Load Specifications at Package Pin table for critical power rails. Altera do not support customers' PCB PDN time domain simulation. It is your responsibility to use the data and specifications in this document along with the simulation guidance in the Board Power Delivery Network Simulations chapter to perform the PDN time domain simulation.

3. Board Power Delivery Network Recommendations

This section outlines recommended designs for power delivery networks, focusing on voltage regulator selection and filtering for power rails fed by a common regulator. It also includes an example of in-house designed development kits and illustrates an example of decoupling capacitor placement on the board.

3.1. Board Decoupling Capacitors and Power Flooding Guides

In addition to on-package decoupling (OPD) (as land-side capacitor (LSC) and die-side capacitor (DSC)), the Agilex 7 device family also offers a cavity site or state to place large size back side capacitors as close as possible to the die or package to improve transient voltage droop response and reduce second or third voltage droop. A total of 15 decoupling capacitors (refer to the Agilex 7 FPGA F-Series, I-Series, and M-Series Decoupling Capacitors Summary table, the bottom side capacitors for VCC core and VCCH) can be added into the Agilex 7 AGF014 Development Kit Board cavity including 9x 0805 47uF (for VCC core) and 6x 0603 22uF (for VCCH) as shown in the Back-side Board Cavity with Large Size Decoupling Capacitors for PCB without Socket, Thin Stackup, and the Use of Micro Via (Agilex 7 AGF014 PCIe Development Kit) figure.

The Back-side Board Cavity with Large Size Decoupling Capacitors for PCB without Socket, Thin Stackup, and the Use of Micro Via (Agilex 7 AGF014 PCIe Development Kit) figure is an example of a decoupling capacitor placement scheme or connection within cavity on the bottom-layer for a PCB designed for the Agilex 7 AGF014 PCIe Development Kit without socket and the use of micro via. The top layer in the Back-side Board Cavity with Large Size Decoupling Capacitors for PCB without Socket, Thin Stackup, and the Use of Micro Via (Agilex 7 AGF014 PCIe Development Kit) figure is assigned for VCC core power and the GND pins on top layer within the decoupling capacitor mounting are connected to the second layer (ground) through a micro via.

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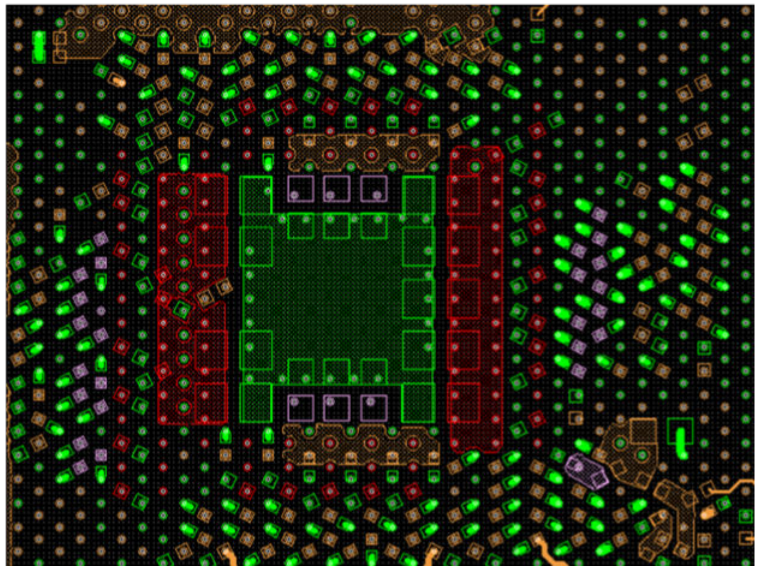
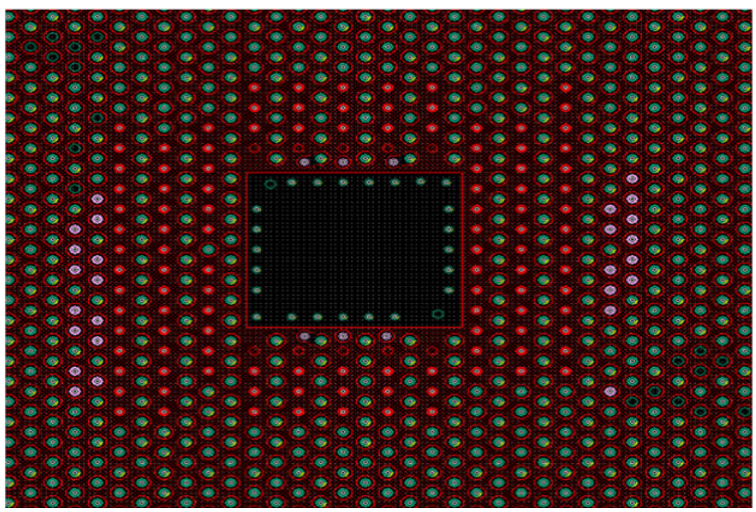
Figure 4. Back-side Board Cavity with Large Size Decoupling Capacitors for PCB without Socket, Thin Stackup, and the Use of Micro Via (Agilex 7 AGF014 PCIe Development Kit)



(a) Bottom-side Decoupling Caps within Cavity Area

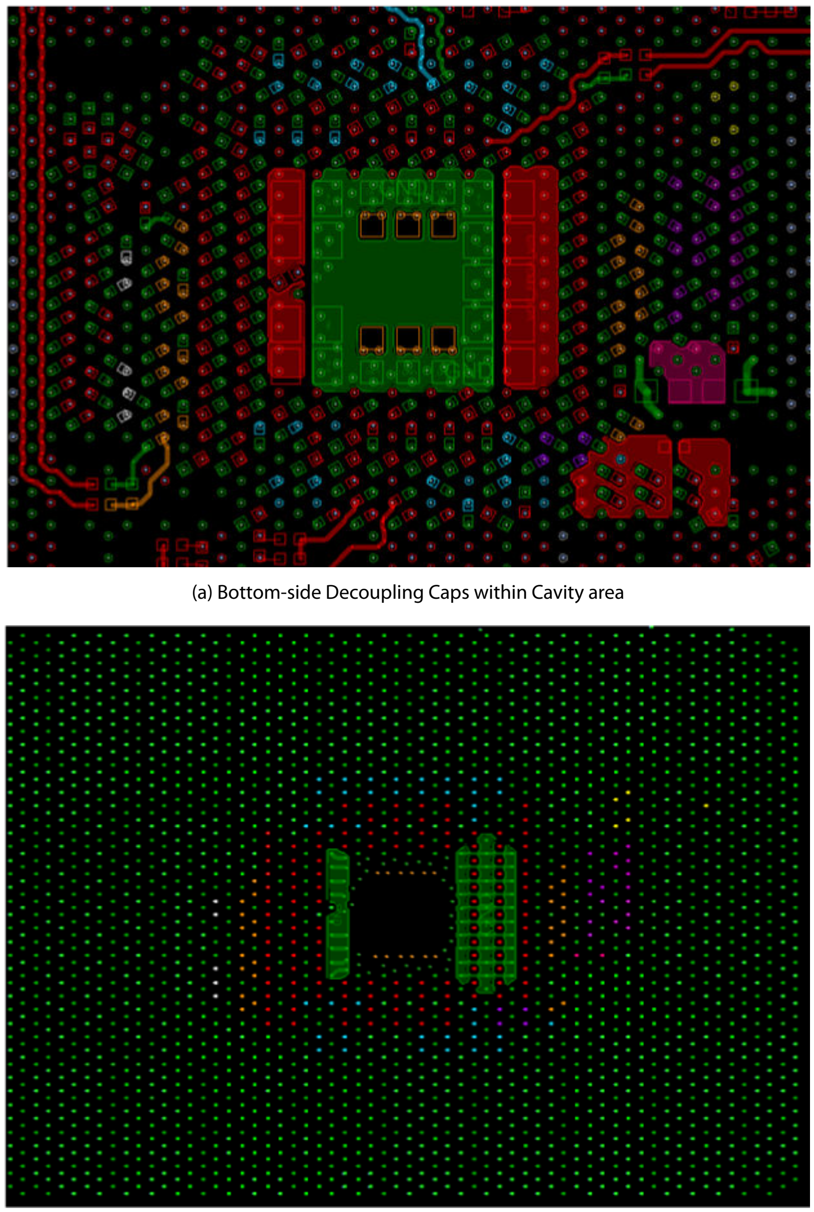


(b) Top-side of PCB within Cavity Area

Because of the location of the cavity capacitors on the bottom side, several GND balls cannot have vias in pad, for stackup without the use of micro via. To ensure that we do not reduce the package current capability, and that we have a low-return inductance path, add a ground island on the top layer connecting those floating balls to adjacent GND vias, as illustrated in the Back-side Board Cavity with Large Size Decoupling Capacitors for PCB without Socket, Thin Stackup, and the Use of Micro Via (Agilex 7 AGF014 PCIe Development Kit) figure.

The Back-side Board Cavity with Large Size Decoupling Capacitors for PCB without Socket, Thin Stackup, and the Use of Micro Via (Agilex 7 AGF014 PCIe Development Kit) figure is an example of decoupling capacitors scheme or connection within the cavity on the bottom layer for a PCB designed of Agilex 7 AGF014 Signal Integrity Development Kit without socket and the use of only through via for GND pins.

Figure 5. Back-side Board Cavity with Large Size Decoupling Capacitors, Thick Stackup, and the Use of Through Via (Agilex 7 AGF014 Signal Integrity Development Kit)



(a) Bottom-side Decoupling Caps within Cavity area

(b) Top-side of PCB within Cavity area

If some of the required decoupling capacitors within the cavity cannot be placed due to some restriction, you can make specification within the pin field to place them as long as they are close to the cavity area.

In addition, other recommended 0201 and 0402 decoupling capacitors can be placed in the via field (FPGA pin field) on bottom layer inside the package shadow. The board side decoupling capacitors (FPGA periphery) recommendation for all rails can be placed either on top layer or bottom layer close to the edge of FPGA device.

This is a summary of the recommended decoupling capacitors placement within cavity for the Agilex 7 AGF012 and AGF014 device family:

- VCC core cavity decoupling capacitors on bottom side:
  - ☐ Thick PCB: 9x 0805 47µF
  - ☐ Thin PCB: 5x 0805 47µF
- VCCH cavity decoupling capacitors on bottom side: 4x 0603 22µF
  - ☐ Option for VCCH: Some 0603 capacitors can be allocated to VCC or VCCIO_PIO/VCCPT depending on power consumption.

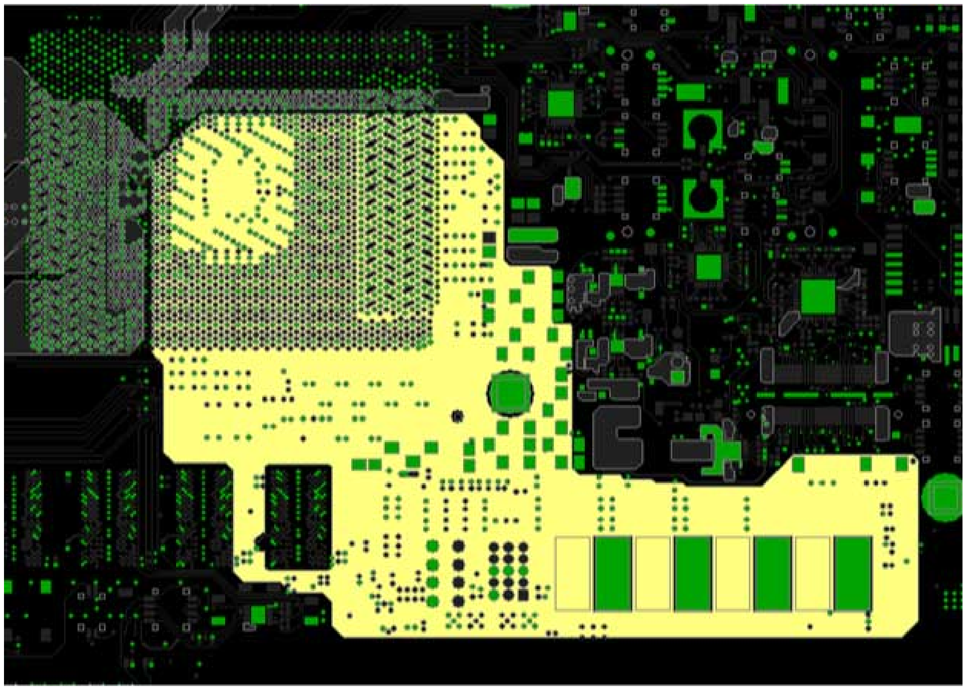
  Note: The power tree and number of decoupling capacitors within the cavity on the Agilex 7 development kit boards may be slightly different than what has been recommended in this application note due to early release of silicon and guideline. The recommended power tree, guideline, and decoupling capacitors in this application note has been well established and validated by measurement for the final device production.

It is due to reliability to have the cavity area on top layer to be free of components due to OPD in package. However, pads or other coppers are allowed in this area on top layer. This means you can place as much capacitors if they fit into the cavity area on the bottom layer connecting capacitors to top layer through via.

Power Flooding

Altera recommends you to include power flood on the top layer of the board for VCCL. The following example is a good design practice that has shown to reduce the loop inductance by 3x due to a shorter path to FPGA fabric and more effective decoupling solutions.

Figure 6. Example of Board Power Flooding



3.2. FPGA Core Fabric VCC Voltage Regulator Selection

Altera recommends you to use the industry standard common footprints for power stages. This provides you the flexibility to choose between different vendors during the validation phases. This also allows you to choose the vendor that best fit your PCB performance and cost targets without impacting the program schedule. The vendors that offer power stages for core in the industry standard common footprints are:

- MPS
- Infineon
- Fairchild
- TI
- Intersil
- ADI

The following controllers have been validated for operation and communication with the Agilex 7 Power Management Controller Tool and added to the existing list of supported controllers and power stages.

The recommended VCC core voltage regulators in the List of Supported Controllers and Power Stages table are available in the Quartus® Prime software for your selection. Altera has already qualified these voltage regulators for the VID application. The PMBus communication is already validated by Altera.

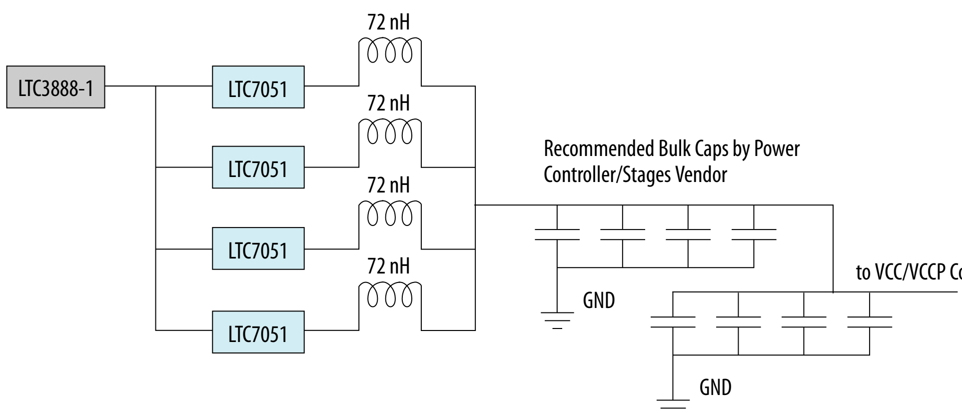
Table 19. List of Supported Controllers and Power Stages

Vendor	Controller	Power Stage	Number of Phase
MPS	MPS2975/2972	MP86956A (60A)	5
Intersil/Renesas	ISL68236	ISL99227 (60A)	5
ADI	LTC3888-1	LTC7051 (60A)	4

Validated VR controllers from the Stratix 10 device are still supported too. For more details, refer to the *Stratix 10 Power Management User Guide*.

The following power design is being tested on internal boards (development kits). Hence, Altera recommends you to follow this VCC core voltage regulator diagram as a reference.

Figure 7. VCC/VCCP Core Voltage Regulator Design Implemented on the Agilex 7 AGF014 2486A FPGA Development Kits



Use of ADI Voltage Regulator

Related Information

- Stratix 10 Power Management User Guide
- SmartVID Debug Checklist and Voltage Regulator Guidelines

3.3. Remote Sense Connections

Die sense pins are provided for the core fabric voltage regulator. The voltage regulator sense line for VCC core must be connected to the differential pair sense lines or pins provided on the package. The voltage regulator feedback inputs shall be connected to this FPGA die remote sense lines.

You are required to use sense lines for Agilex 7 core, including the VID and multi-voltage designs. You are also required to add sense line to the F-Tile transceivers power rail design because of tight specifications. VCCERT1_FHT_GXF power rails have dedicated sense pins at package level.

Note: For those power rails which do not have dedicated sense pins at package level, the IR drop can still be compensated by using a voltage regulator with the sense feedback pin. To find the sense pins, you must do the IR drop analysis along with plotting power/current distribution of that specific power rail at package level to find the pins which are located further from the voltage regulator and have the maximum IR drop.

Note: Any sense line used on the board for that specific power rail must be placed before LC filter. If a sense line is placed after LC filter, it can result in oscillation to the rails and an unstable voltage that is an input to the voltage regulator feedback pin function. If the power rail is using LDO, LC filter is not required.

3.4. Load Line Requirements

Load line is optional for Agilex 7 device family packages. The load line is required by some workloads that may need it.

3.5. VCC Core Board Current Slew Rate

The VCC core fabric has a very high current slew rate. But some of that is filtered out by the metal-insulator-metal (MIM) capacitors as they provide the lowest impedance to the load because of their proximity. With addition of OPDs, most of the remaining fast edges are filtered out, leaving the board cavity capacitors and voltage regulator capacitors to handle only a fraction of the die level current.

You must ensure the PDN can deliver the current specified at the package balls. The Board LC Recommended Filters for Noise Reduction in Combined Power Delivery Rails section describes the recommended PCB system-level simulation to ensure the voltage tolerance or specification at package balls are met through the design.

4. Board LC Recommended Filters for Noise Reduction in Combined Power Delivery Rails

The part numbers of ferrite bead (FB) and inductances (L) selected in this chapter are the parts which are recommended by Altera. However, you can select different part numbers as long as the inductance/impedance at certain frequency, resistance, and other parameters of the new parts are similar to recommended FB/L part (similar inductance/impedance and similar to lower resistance within frequency range mentioned in the data sheet of recommended part).

Caution: Any sense line used on board must be placed before LC filter. If the sense line is placed after the LC filter, this results in a serious oscillation to the rails and unstable voltage that is an input to the voltage regulator feedback pin function.

Note: If the power rail is using LDO, you do not need to use the LC filter.

If you are using the recommended switching voltage regulator (LTC7151S), with a proper design from ADI, and meet the simulated 5 mV ripple specification, you do not need to use the LC filter for VCCERT_FGT_GXF, VCCERT1_FHT_GXF, and VCCRT_GXR rails.

Table 20. Recommended Inductors and Ferrite Beads for the LC Filter

Part MPN	L (nH)	DCR (mΩ)
SL1616A-R10MHF	100	0.32
SL1616A-R05MHF	50	0.32
BLM18SN220TZ1	—	4
BLM18KG260TN1	—	7

The inductors or ferrite beads listed in the table above have been validated on the Altera development kits and should be used as a reference only. You can choose other inductors or ferrite beads according to specific applications, as long as the specifications stated above are met and the LC filter works well. Altera recommends you to run the simulation to ensure there is no further changes.

Table 21. Recommended Capacitors Used for the LC Filter

Part MPN	Size (mm/inch)	Temperature Character	Capacitor Value (µF)	Capacitor Tolerance (%)	Rated Voltage (V)
GRM21BC80G107ME15	2012M/0805	X6S	100	±20	4
GRM21BC80G476ME15	2012M/0805	X6S	47	±20	4
GRM21BC81C226ME44	2012M/0805	X6S	22	±20	16
GRM188C80G476ME01	1608M/0603	X6S	47	±20	4

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The capacitors listed in the table above have been validated on the Altera development kits and should be used as a reference only. You can choose other capacitors according to specific applications, package size used, temperature range, form factor, and rated voltage, as long as the specifications stated above are met and the LC filter works well. Altera recommends you to run the simulation to ensure there is no further changes.

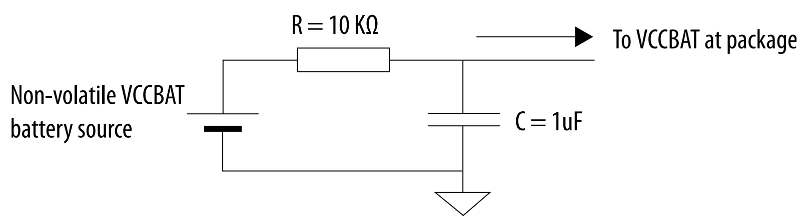
Choosing the LC Filter based on the DCR Value

DC drop specification must be around 0.5% of the voltage supply. If the voltage rail is 0.9 V, current is 1 A, BLM18SN220TZ1 DCR is 4 mΩ, DC drop = current x DCR = 1 A x 4 mΩ = 4 mV. The 0.5% specification of 0.9 V is 4.5 mV, which is within the specification. For a higher current value, SL1616 is the most suitable as it has lowest DCR.

VCCBAT RC Circuit Using the Device Security AES BBRAM Key

The connection shown below is recommended for the VCCBAT. Connect this pin to a non-volatile battery power source in the range of 1.0 V to 1.8 V. A series RC (R=10 KΩ, C=1 µF) circuit must be added to the VCCBAT rail.

Figure 8. VCCBAT Rail RC Circuit Recommendation



P1V8_GR2 (Agilex 7 F-Series and I-Series)/P1V8_GR3 (Agilex 7 M-Series) Filters Recommendation for Device Fabric

The connections below are recommended for the P1V8_GR2_FLTR voltage rails for the Agilex 7 F-Series and I-Series devices or the P1V8_GR3_FLTR voltage rails for the Agilex 7 M-Series devices for noise filtering. The P1V8_GR2_FLTR for the Agilex 7 F-Series and I-Series devices or the P1V8_GR3_FLTR for the Agilex 7 M-Series devices consists of VCCADC, VCCPLL_SDM, and VCCPLL_HPS.

Figure 9. VCCADC, VCCPLL_SDM, and VCCPLL_HPS Filter Recommendation

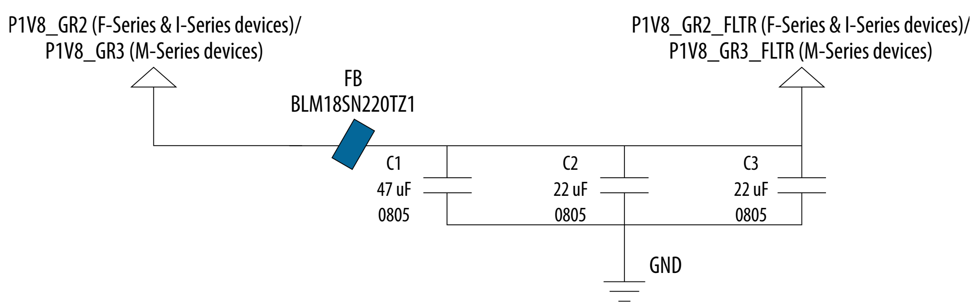
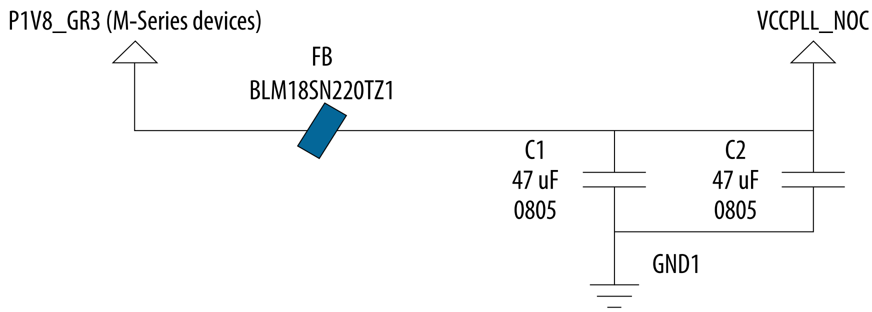


Figure 10. VCCPLL_NOC LC Filter Recommendation



P1V2_GR3 Filters Recommendation for Device Fabric

The following figure shows the filter recommendation for VCCA_PLL in 1.2 V VCCIO power rail.

Figure 11. VCCA_PLL Filter Recommendation

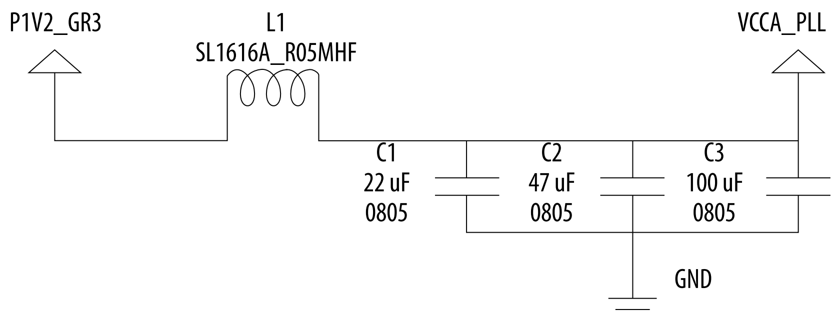
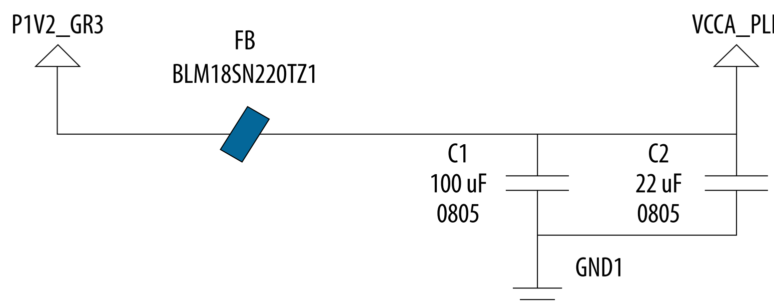


Figure 12. VCCA_PLL LC Filter Recommendation (Low Power Scenario)



P0V8_GR1 Filters Recommendation for Device Fabric

The following figure shows the filter recommendation for the VCCPLLDIG_SDM, VCCPLL_NOC, and VCCPLLDIG_NOC in P0V8_GR1 power rail.

Figure 13. VCCPLLDIG_SDM Filter Recommendation

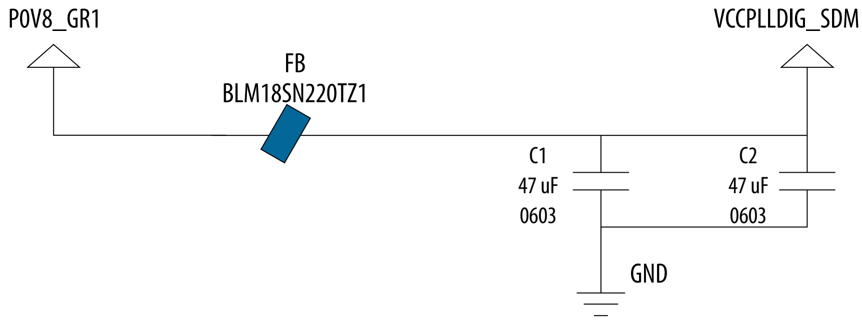


Figure 14. VCCLPLL_NOC LC Filter Recommendation

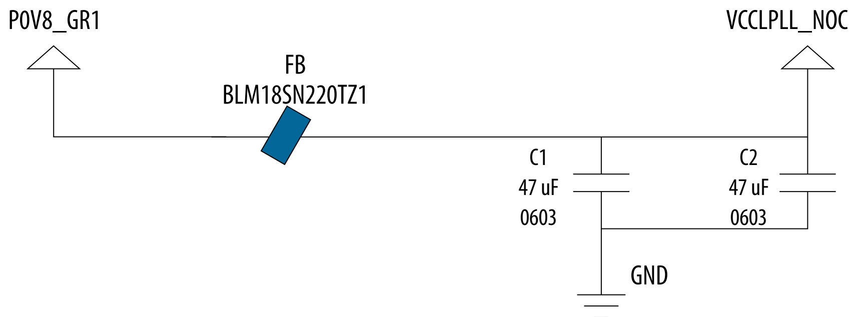
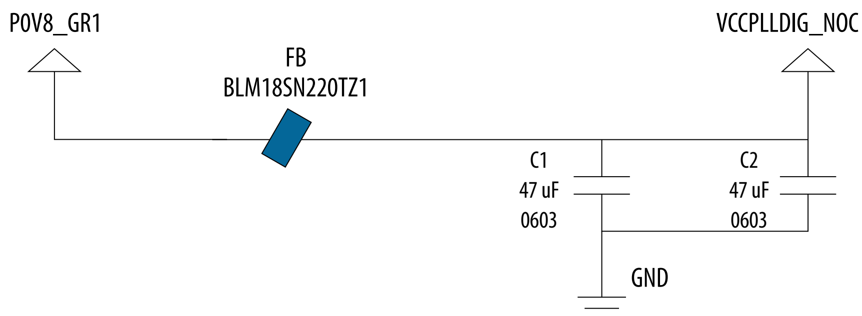


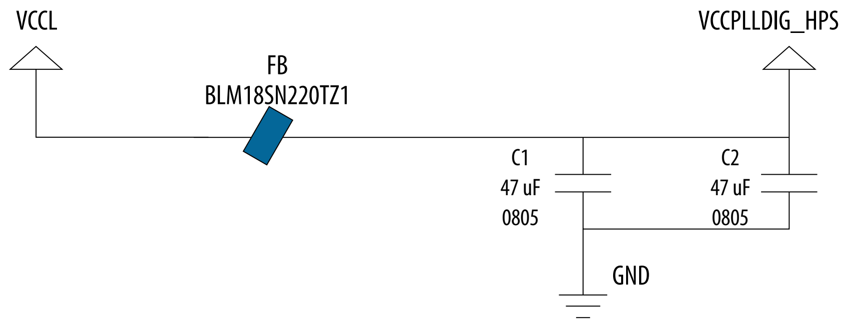
Figure 15. VCCPLLDIG_NOC LC Filter Recommendation



VCCL Filters Recommendation for Device Fabric

The following figure shows the filter recommendation for the VCCPLLDIG_HPS in VCCL power rail.

Figure 16. VCCPLLDIG_HPS Filter Recommendation



4.1. P-Tile Rail LC Filter Board Scheme and Connection

The filtering topology in the Filter Recommendation for VCCCLK_GXP per Tile, Filter Recommendation for VCCH_GXP per Tile, and Filter Recommendation for VCCRT_GXP per Tile figures are recommended for the P-Tile voltage rails (VCCCLK_GXP, VCCH_GXP, VCCRT_GXP) in the Power Tree section for noise filtering purpose. The filter can be placed as close to FPGA as periphery capacitors in the recommended decoupling capacitors table in the Decoupling Capacitors Recommendation section. In addition to the LC filter, you must also add the bottom or backside capacitors recommended in the decoupling capacitors table in the Decoupling Capacitors Recommendation section within pin or via field on the bottom layer. Implement one set of LC filters for each tile on the FPGA.

Figure 17. Filter Recommendation for VCCCLK_GXP per Tile

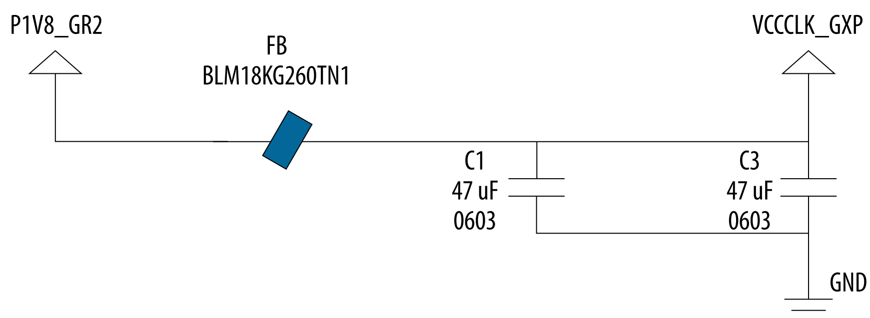


Figure 18. Filter Recommendation for VCCH_GXP per Tile

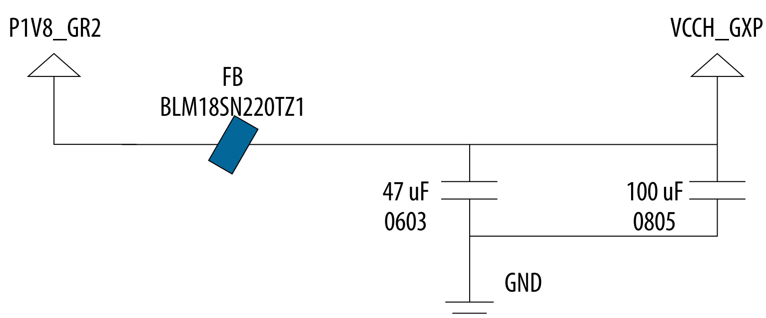


Figure 19. Filter Recommendation for VCCH_GXP per Tile (Low Power Scenario)

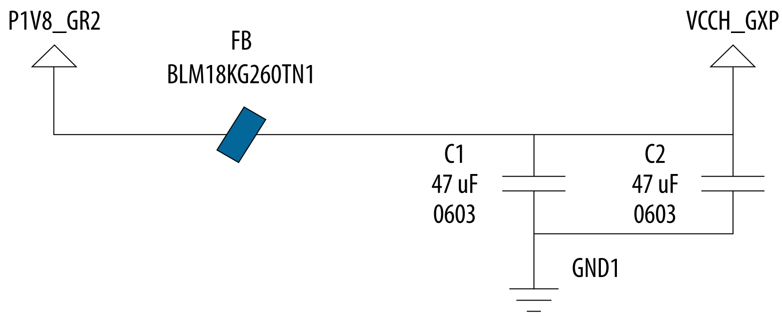


Figure 20. Filter Recommendation for VCCRT_GXP per Tile

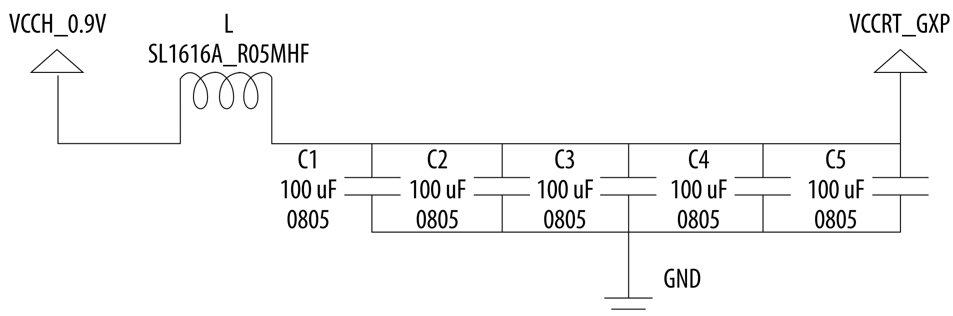
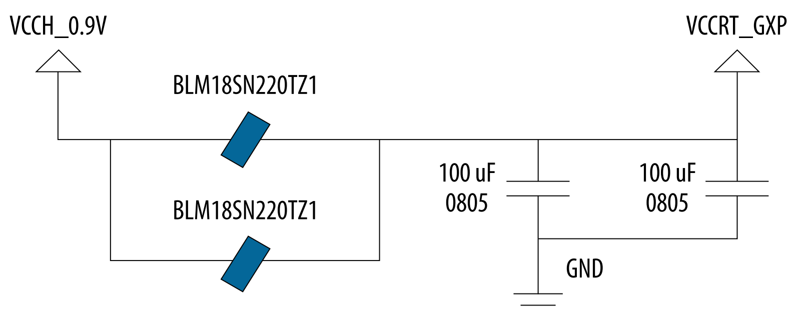


Figure 21. Filter Recommendation for VCCRT_GXP per Tile (Low Power Scenario)



4.2. E-Tile Rail LC Filter Board Scheme and Connection

The E-Tile has a strict connection requirement to eliminate the need for power-down sequencing (PDS) control circuits. With the connections in the Connection Requirement and Filter Recommendation for VCCRT_GXE per Tile and Connection Requirement and Filter Recommendation for VCCRTPLL_GXE per Tile figures, the E-Tile eliminates the PDS requirement existing in the Stratix 10 device family—no pull-down discharge FETs or resistors on voltage rails are required. The filter can be placed as close to FPGA as periphery capacitors in the Agilex 7 FPGA F-Series, I-Series, and M-Series Decoupling Capacitors Summary table. Implement one set of LC filters for each tile on the FPGA.

Figure 22. Connection Requirement and Filter Recommendation for VCCRT_GXE per Tile

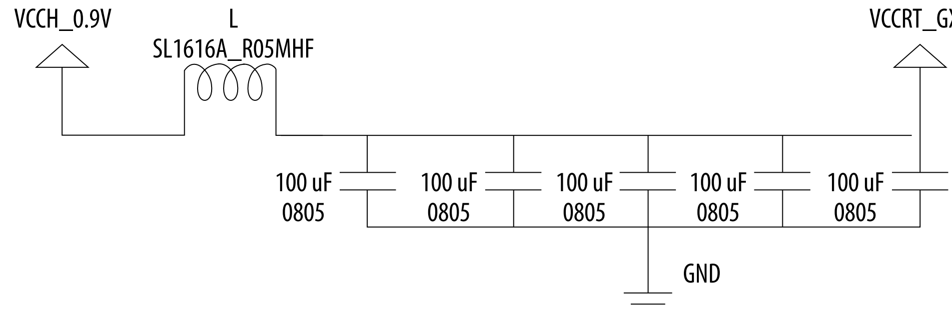


Figure 23. Connection Requirement and Filter Recommendation for VCCRT_GXE per Tile (Low Power Scenario)

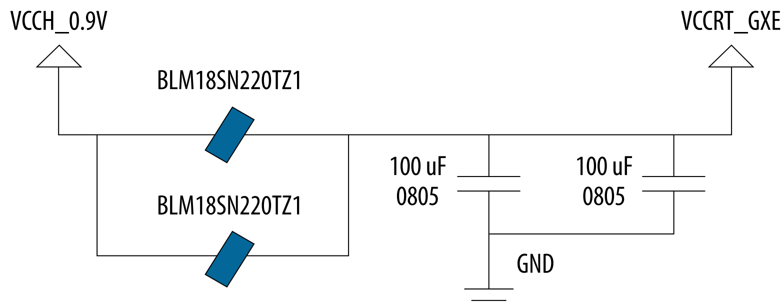
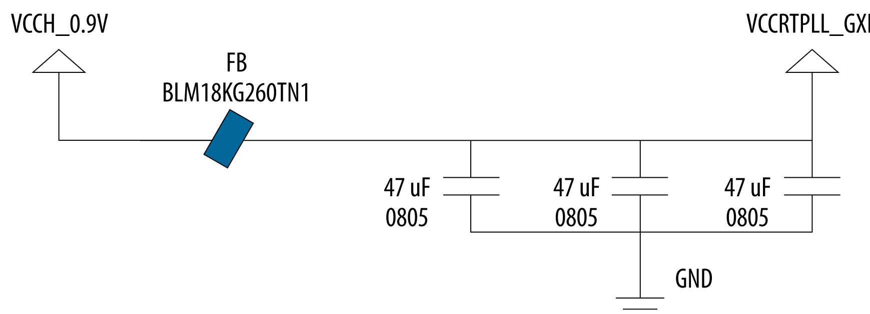


Figure 24. Connection Requirement and Filter Recommendation for VCCRTPLL_GXE per Tile



4.3. F-Tile Rail Board Connection and LC Filter Recommendation under Noisy Voltage Regulator

The LC filter recommendation in this chapter is not required for the F-Tile power rails if you meet the voltage regulator (VR) ripple requirement in the Maximum Ripple Requirements for the F-Tile Analog Power Rails table. Altera recommends using the LTC7151S voltage regulator to meet very tight noise specifications for VCCERT1_FHT_GXF and VCCERT2_FHT_GXF. You can use any other voltage regulators for the F-Tile power rails as long as you meet the voltage regulator ripple specifications by selecting a proper voltage regulator. If you do not meet the VR ripple specification in the Maximum Ripple Requirements for the F-Tile Analog Power Rails table by using a noisy voltage regulator, Altera recommends using the LC filters in this chapter to block the high noise. Implement one set of LC filters for each tile on the FPGA.

Table 22. Maximum Ripple Requirements for the F-Tile Analog Power Rails

Power Rail	Maximum Ripple (mVp-p)
VCCERT_FGT_GXF	5
VCCERT1_FHT_GXF	5
VCCERT2_FHT_GXF	5
VCCH_FGT_GXF	7
VCCEHT_FHT_GXF	7

Figure 25. Filter Recommendation for VCCH_FGT_GXF per Tile

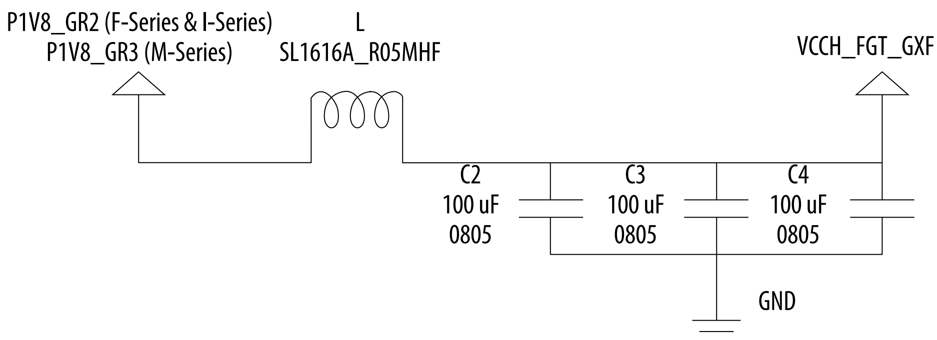
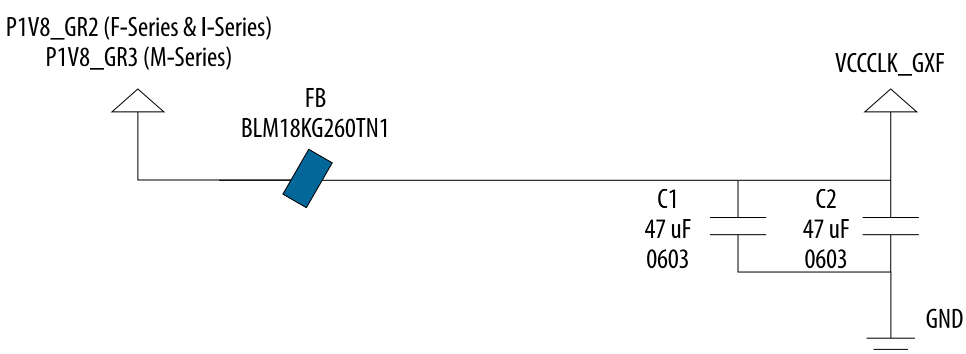


Figure 26. Filter Recommendation for VCCCLK_GXF per Tile



4.4. R-Tile Rail Board Connection and LC Filter Recommendation under Noisy Voltage Regulator

The R-Tile requires a strict connection requirement to eliminate the need for power-down sequencing (PDS) control circuits. Using the board connections shown below, the R-Tile removes the PDS requirement (such as the PDS requirement in the Stratix 10 device family)—no pull-down discharge FETs or resistors on voltage rails is required. Therefore, the Agilex 7 device is a PDS-free FPGA, except for E-Tile.

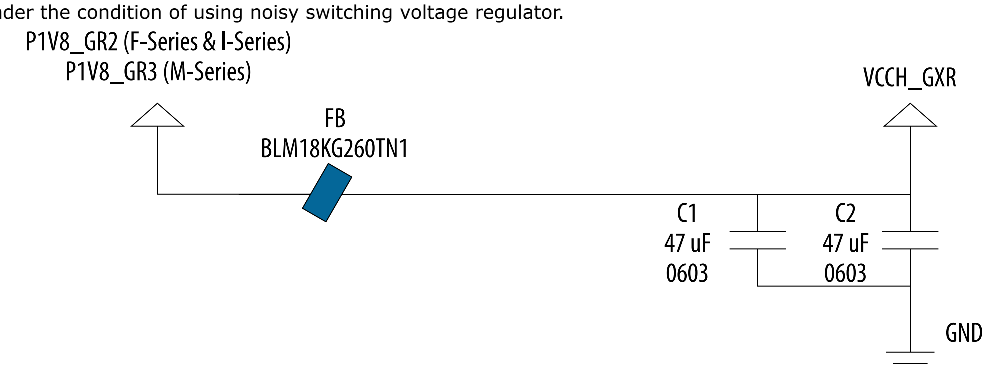
For the VCCRT_GXR rail that is designed with the recommended LTC7151S voltage regulator (500 KHZ switching frequency is recommended), with a very tight noise specification, you do not need to add filter to this power rail (refer to the recommended power tree). However, if noisy switching voltage regulator is used, an elaborate filter design is required to dampen the noise in the power rail. Perform the PDN transient simulation to validate this power rail design against the specifications.

If you are using the LDO voltage regulator to meet the tight noise specification for this power rail, the recommended LDO is 2x ADP1765 and it is not needed to add LC filter when using LDO.

The maximum voltage regulator (VR) voltage ripple allowed for the VCCRT_GXR is 5 mV and for VCCH_GXR is 7 mV. If you do not meet this VR ripple specification for the R-Tile power rails by selecting a noisy VR, Altera recommends using the recommended LC filters to block the noise.

Figure 27. Connection Requirement and Filter Recommendation for VCCH_GXR per Tile

Under the condition of using noisy switching voltage regulator.



5. PCB PDN Design Guideline for Unused Tiles

This chapter describes the PDN design guidelines for unused tiles on PCB design. For more information, refer to the *Agilex 7 Device Family Pin Connection Guidelines*.

Related Information

- Agilex 7 Device Family Pin Connection Guidelines: F-Series and I-Series
- Agilex 7 Device Family Pin Connection Guidelines: M-Series

5.1. PDN Design Guideline for Unused E-Tile

All E-Tile power rails (ending with _GXE) must be powered on PCB even if the E-Tile is not used in your board design. However, due to only small static or bypass current usage, the recommended decoupling capacitors in the Agilex 7 E-Tile Decoupling Capacitors Summary per Tile table for E-Tile power rails in this document are not required on PCB. You must still meet the voltage specification at package pin for the E-Tile power rails by selecting a proper voltage regulator (such as low ripple, DC setpoint, recommended VR bulk capacitors, etc.) and considering the static or bypass current.

5.2. PDN Design Guideline for Unused P-Tile

All P-Tile power rails (ending with _GXP) must be powered on PCB even if the P-Tile is not used in your board design. However, due to only small static or bypass current usage, the recommended decoupling capacitors in the Agilex 7 P-Tile Decoupling Capacitors Summary per Tile table for the P-Tile power rails in this document are not required on PCB. You must still meet the voltage specification at package pin for the P-Tile power rails by selecting a proper voltage regulator (such as low ripple, DC setpoint, recommended VR bulk capacitors, etc) and considering the static or bypass current.

5.3. PDN Design Guideline for Unused R-Tile

If R-Tile is not used on your PCB design, all the R-Tile power rails (ending with _GXR: VCC_HSSI_GXR, VCCH_GXR, VCCED_GXR, VCCRT_GXR, VCCE_DTS_GXR, VCCE_PLL_GXR, VCCHFUSE_GXR, and VCCCLK_GXR) can be connected to GND. You must disconnect them from the power tree. VCCH in the power tree must still be powered on using the recommended decoupling capacitors listed in this document.

5.4. PDN Design Guideline for Unused F-Tile

If the entire F-Tile is not used on your PCB design, all F-Tile power rails (ending with _GXF) can be connected to GND. You must disconnect them from the power tree. VCCH in the power tree must still be powered on with the recommended decoupling capacitors listed in this document.

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If only the high-speed 116G FHT transceivers/channels are not used in the F-Tile, the F-Tile power rails (ending with _FHT_GXF) in the power tree can be connected to GND. All other power rails remaining in F-Tile must be powered on according to the specification by using the recommended decoupling capacitors listed in this document.

6. PCB Voltage Regulator Recommendation for PCB Power Rails

Altera recommends the Agilex 7 PCB-based designs to use the recommended voltage regulators because of its capability, efficiency, and performance which have been validated on various Agilex 7 device family boards. Although you can use other voltage regulators, you are advised to use tested and trusted power solutions to remove the burden of validating other vendor solutions. This is to ensure that you can focus your bandwidth on validating and optimizing the FPGA intrinsic performances.

You can use other voltage regulators as long as you meet the PCB power rail tolerance and specification in the Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables measured at package pin. Altera does not support customers' PCB PDN time domain simulations. It is your responsibility to use the data and specifications in this document along with the guidance in the Board Power Delivery Network Simulations chapter to perform PDN time domain simulation.

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7. Board Power Delivery Network Simulations

In this section, the PDN post-layout simulation is shown in the Methodology for Device PDN and Transient Noise Analysis figure for any Agilex 7 device family board design and system-level PDN simulation.

Figure 28. Methodology for Device PDN and Transient Noise Analysis

Step load at the package pin is injected to the PCB model to meet voltage droop (DC + AC) at the package pin.

Altera recommends you to follow the above-mentioned guidelines to design all power rails on the PCB with the recommended decoupling capacitors, voltage regulators, and LC filtering. In the post-layout phase, Altera recommends you to do the IR drop and transient (time domain) PDN analysis for PCB only. This means, unconventionally, Altera do not recommend impedance target and frequency target analysis (frequency domain simulation) for the Agilex 7 device.

To ensure the PDN design performance is within the required tolerance or specification in the Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables, time domain post-layout PDN simulation for some critical power nets such as VCC core, VCCP, VCCPT, VCCIO_PIO, VCCH, and power rails for E-Tile, P-Tile, F-Tile, and R-Tile must be performed.

PDN time domain simulation is only performed on PCB from voltage regulator to package ball. Therefore, package, OPDs, and on-chip models are not required for the PDN time domain simulation.

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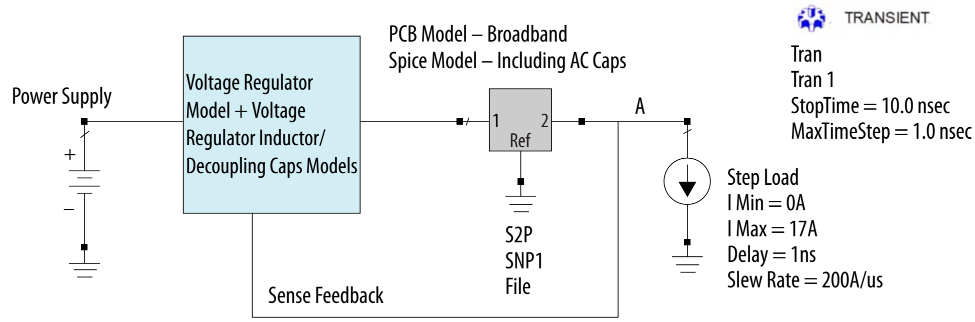
The following steps show the time domain PDN simulation (as shown in the Time Domain PDN Test Bench Example for Agilex 7 AGF014 VCC Core figure):

1. Obtain the implemented VRM SPICE model for the target power rail.
2. Extract post-layout PCB model (HSPICE or scattering parameters by using tools such as PowerSI) of the PCB with decoupling capacitors and LC filtering from the voltage regulator (including VRM recommended bulk decoupling capacitors by vendor) to package pin (if use of scattering parameters, the PCB model shall be extracted from DC up to 1 GHz). Altera recommends you to convert scattering parameters to circuit model by use of any broadband Spice or IDEM tool to avoid problematic simulation. To avoid simulation divergence, you shall include the small to medium decoupling capacitors on PCB extraction and define ports for the large and bulk capacitors on the PCB when extracting its model. Then, you add the large/bulk capacitors (in the format of spice models) externally in the schematic (as explained in step 3).
3. Build a schematic in any possible EDA tool (Keysight ADS or Cadence or LTspice or Simplis and so forth) with the voltage regulator model (possible HSPICE model) and PCB model extracted from previous step.
    - This schematic represents the voltage regulator plus the PCB or decoupling capacitors model up to package pins.
    - Package, OPDs, or die model are not built into this schematic (Step load at package pin covers frequencies for only PCB, which means high frequency current components are eliminated through package and on-die).
    - Connect the sense pins from the package pin feedback to the voltage regulator sense pins.
4. Connect the maximum step load current at the package pins shown in the Agilex 7 Device Family Transient and Step Load Specifications at Package Pin table (for example, for Agilex 7 AGF014 core, 200A/µs slew rate and step load of 17A).
5. Probe voltage drop at the package pin to see if the power rail specification in the Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables is met (for example, for VCC core, the DC+AC voltage tolerance is ±3%).
    - If not meeting the package power rail tolerance or specification in the Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables, you must check the PCB and adjust the decoupling capacitors or locations.

Figure 29. Time Domain PDN Test Bench Example for Agilex 7 AGF014 VCC Core

"A" is the VCC node at the package ball (all VCC pins at the package are connected to A). Voltage at "A" must be evaluated based on the voltage tolerance.



You must notice that the Time Domain PDN Test Bench Example for Agilex 7 AGF014 VCC Core figure shows a simplified schematic for the PDN transient simulation. In order to avoid non-convergence condition in TD simulation, Altera recommends you to only include small decoupling capacitors in the PCB model extraction and define ports for large/bulk decoupling capacitors at PCB level and add them to the schematic in the Time Domain PDN Test Bench Example for Agilex 7 AGF014 VCC Core figure manually.

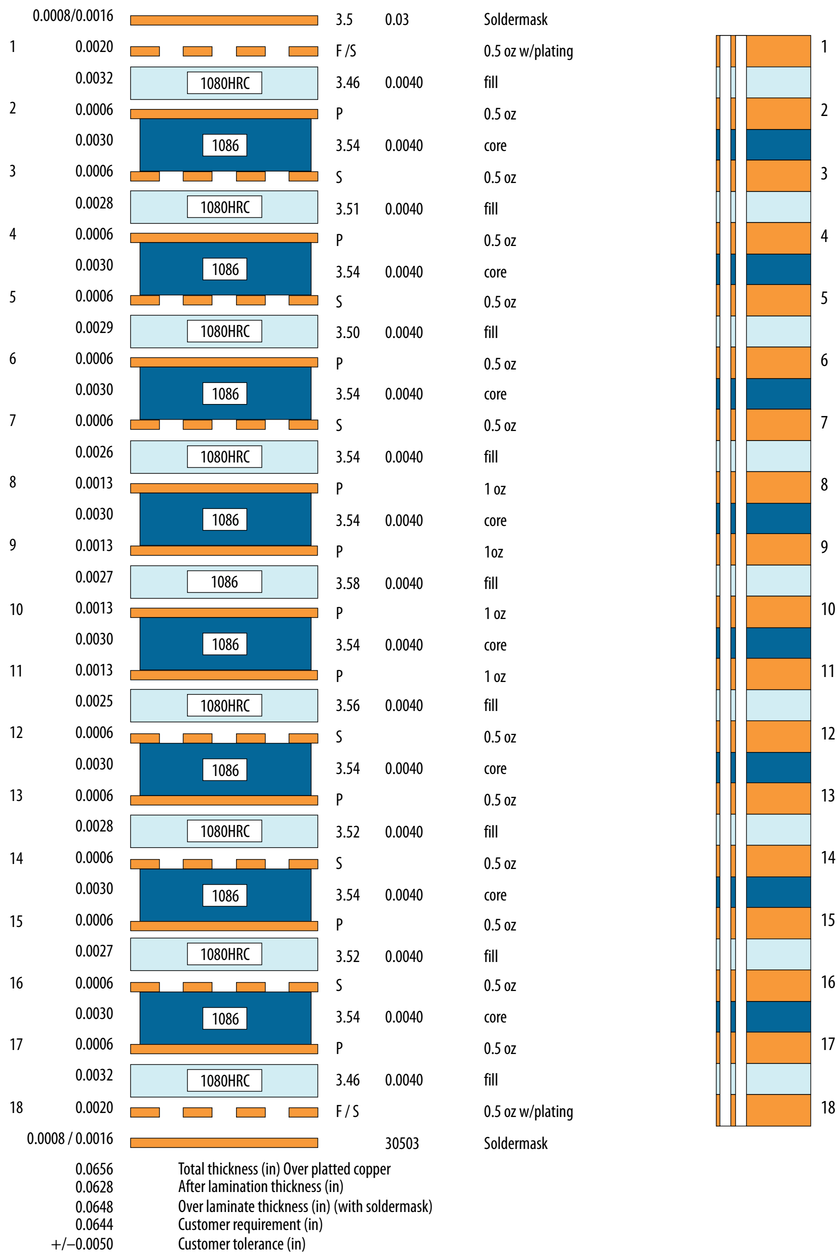
Some of the PCB power rails in the recommended power tree in the Recommended Agilex 7 AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices and Recommended Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon figures are built by merging various power rails (with or without filter). You must define ports in the PCB modeling for each power rail to be able to inject the respective step load for that power rail in the PDN simulation schematic.

The recommended step load along with the static current (obtained from PTC) in a format of a pulse for each power rail is added to power rail ports in the PDN simulation schematic (e.g., Time Domain PDN Test Bench Example for Agilex 7 AGF014 VCC Core figure) and the voltage droop and overshoot are measured against the specification listed in Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables.

The PDN IR drop analysis is a DC simulation and must be performed on all power rails on the PCB up to package pins to meet the electrical specifications listed in the respective *Agilex 7 FPGAs and SoCs Device Data Sheets*.

The Reference Stackup figure shows the reference stackup used in the PDN design guideline and FPGA decoupling capacitors extraction. However, the FPGA PDN performance is also validated with a thicker PCB such the DK-SI-AGF014E3ES board designed in-house.

Figure 30. Reference Stackup



Related Information

- Agilex 7 FPGAs and SoCs Device Data Sheet: F-Series and I-Series
- Agilex 7 FPGAs and SoCs Device Data Sheet: M-Series

8. Agilex 7 Device Family PDN Design Summary

The summary of the Agilex 7 device family PDN design guidelines are as follow:

1. Current PDN design guidelines stand for the maximum power consumption—the worst use case.
    - If for any reason (various applications, configurations, or the PTC) power data is lower than the maximum power used for the PDN design guideline, you must scale the recommended decoupling capacitors based on the ratio of design current to the maximum current. Use of ratio is an estimate and time domain simulation are mandatory to ensure meeting package ball voltage specification.
2. Apply the recommended power-up or power-down sequence grouping on the PCB. For more information, refer to *AN 692: Power Sequencing Considerations for Cyclone® 10 GX, Arria 10, Stratix 10, and Agilex 7 Devices* and *Agilex 7 Power Management User Guide*.
3. Use the recommended power tree presented in the *Power Tree* section for each Agilex 7 device with the suggested merged power nets.
    - For example, minimum 9 x voltage regulator is required on the PCB for the Agilex 7 AGF014 2486A Package Early Silicon and minimum 10 x voltage regulator is required on the PCB for Agilex 7 AGF014 production silicon. The recommended voltage regulators are only for FPGA and do not cover other devices on the board.
    - The minimum recommended number of voltage regulators on PCB is due to cost, area, and power-effective solution strategies. However, you can separate all power rails by the use of separate voltage regulators.
4. Use the recommended voltage regulators in the power tree or design your own voltage regulator based on the required maximum ripple or total current support per power rail on the PCB-VRM inductors or bulk capacitors must be designed separately. Tables in the *Decoupling Capacitors Recommendation* section show the FPGA decoupling capacitors and do not include the voltage regulator bulk capacitors.
5. Use the recommended bottom-side or FPGA periphery decoupling capacitors for each power net.
6. Use the recommended LC filters for power nets.
7. Use of sense line for IR drop compensation.
8. Configure the FPGA to follow the maximum recommended step load allowed at the package pin.

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9. Do post-layout simulation for the IR drop analysis to see if this is within the DC specification at the package pin in the Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables.
10. Do post-layout time domain PCB simulation up to the package pin for critical power nets such as the VCC core to meet the AC voltage tolerance or specification at the package pin in the Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance, Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance, and Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance tables.
11. If not meeting the voltage tolerance (DC or AC) at the FPGA package pin, you must check the PCB and update the decoupling capacitors and redo the simulations.

Related Information

- Power Tree on page 7
- Decoupling Capacitors Recommendation on page 24
- AN 692: Power Sequencing Considerations for Cyclone 10 GX, Arria 10, Stratix 10, and Agilex 7 Devices
- Agilex 7 Power Management User Guide

9. Document Revision History for the Agilex 7 Power Distribution Network Design Guidelines

Document Version	Changes
2025.08.18	• Updated Table: *Agilex 7 F-Series, I-Series, and M-Series Decoupling Capacitors Summary* to update the Device and add R24D package. • Updated the device guideline status for F-Tile and R-Tile in *Device Guidelines Status for Agilex 7 Devices* from *Preliminary* to *Final*.
2025.04.07	Updated the device guideline status for the following devices from *Advance* to *Preliminary*: • AGM 032/039 (R47A) • AGM 032/039 (R31B) • AGM 032/039 (R47B)
2025.01.22	• Updated *Agilex 7 Power Distribution Network Design Guidelines Overview*. • Updated the device guideline status for AGI 041 (R31E) devices from *Advance* to *Preliminary*. • Updated *Power Tree* for clarity. • Updated the P1V2_GR3b power rail tolerance in Table: *Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance*. • Updated *Board Power Delivery Network Simulations*.
2024.10.18	• Updated Table: *Agilex 7 F-Tile Power Rail Nets and Subsystem Details*: ☐ Corrected rail name VCCLK_GXF to VCCCLK_GXF. ☐ Updated system connection description for VCCCLK_GXF. • Made minor editorial edits to the document.
2024.08.09	Updated the following figures: • Figure: *Recommended Agilex 7 AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices* • Figure: *Recommended Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon* • Figure: *Recommended Agilex 7 AGM (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Engineering Sample (ES) Silicon*.
2024.07.08	• Renamed the document title to *Agilex 7 Power Distribution Network Design Guidelines*. • Updated Table: *Device Guidelines Status for Agilex 7 Devices (F-Series)*: ☐ Added AGF 006/008/012/014 (R24D). ☐ Updated the device guideline status for the following devices from *Advance* to *Preliminary*: • AGF 006/008 (R16A) • AGF 006/008/012/014/019/022/023/027 (R24C) • AGF 019/022/023/027 (R31C) • Updated Table: *Device Guidelines Status for Agilex 7 Devices (I-Series)*: ☐ Added AGI 041 (R31E). ☐ Updated the device guideline status for the following devices from *Advance* to *Preliminary*: • AGI 022/027 (R31A) • AGI 041 (R29D) • AGI 041 (R31B)

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Document Version	Changes
	• Updated Table: *Device Guidelines Status for Agilex 7 Devices (M-Series)*: ☐ Added AGM 039 (R47B). ☐ Corrected device name for R31B package to AGM 039 only. • Removed Table: *Agilex 7 Device PCB Power Rail Tolerance*. • Added the following tables: ☐ Table: *Agilex 7 AGF Devices with only E-Tile and P-Tile PCB Power Rail Tolerance* ☐ Table: *Agilex 7 AGF and AGI Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance* ☐ Table: *Agilex 7 AGM Devices with only F-Tile, or both F-Tile and R-Tile PCB Power Rail Tolerance* ☐ Table: *Agilex 7 F-Series, I-Series, and M-Series Decoupling Capacitors Summary* ☐ Table: *Agilex 7 Device Family Transient and Step Load Specifications at Package Pin* • Updated the following figures: ☐ Figure: *Recommended Agilex 7 AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices* ☐ Figure: *Recommended Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon* ☐ Figure: *Recommended Agilex 7 AGM (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Engineering Sample (ES) Silicon*. • Updated the system connections for VCCH_GXP and VCCCLK_GXP in Table: *Agilex 7 P-Tile Power Rail Nets and Subsystem Details*. • Updated *Load Line Requirements*. • Made editorial edits throughout the document.
2023.12.04	• Added the R31B package support in Table: *Device Guidelines Status for Agilex 7 Devices (M-Series)*. • Updated the AGM039/AGM032 device package support in Table: *Agilex 7 Device Family Transient and Step Load Specifications at Package Pin*. • Updated the AGM039/AGM032 device package support in Table: *Agilex 7 FPGA F-Series, I-Series, and M-Series Decoupling Capacitors Summary*.
2023.09.29	• Updated the *Power Nets and Transient Specifications* section. • Updated the *Agilex 7 FPGA Packages Board-Level Decoupling Capacitors Summary* section. • Updated the *Agilex 7 R-Tile Board-Level Decoupling Capacitors Summary* section. • Updated the *Board LC Recommended Filters for Noise Reduction in Combined Power Delivery Rails* section. • Updated Table: *Device Guidelines Status for Agilex 7 Devices (F-Series)*. • Updated Table: *Device Guidelines Status for Agilex 7 Devices (I-Series)*. • Updated Table: *Agilex 7 R-Tile Power Rail Nets and Subsystem Details*. • Updated Table: *Agilex 7 F-Tile Power Rail Nets and Subsystem Details*. • Updated Table: *Agilex 7 Device Family Transient and Step Load Specifications at Package Pin*. • Updated Table: *Agilex 7 FPGA F-Series, I-Series, and M-Series Decoupling Capacitors Summary*. • Updated Table: *Recommended Capacitors Used for the LC Filter*. • Updated Figure: *Recommended Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon*. • Updated the figure title in Figure: *Connection Requirement and Filter Recommendation for VCCH_GXR per Tile*.
2023.06.26	• Added reference to *AN 974: Stratix 10 and Agilex 7 SmartVID Debug Checklist and Voltage Regulator Guidelines*. • Updated the *Agilex 7 FPGA Packages Board-Level Decoupling Capacitors Summary* section to include AGI041 device. • Updated Table: *Device Guidelines Status for Agilex 7 Devices (I-Series)* to include AGI 041 R29D and R31B packages. • Updated Table: *Agilex 7 Device Family Transient and Step Load Specifications at Package Pin* to include AGI041 device. • Updated Table: *Agilex 7 FPGA F-Series, I-Series, and M-Series Decoupling Capacitors Summary* to include AGI041 R29D and R31B packages.

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Document Version	Changes
2023.04.03	• Added support for the Intel Agilex® 7 M-Series devices. • Added *Recommended Power Tree for the Intel Agilex 7 AGM (with only F-Tile, or both F-Tile and R-Tile) Device Packages* section. • Added *P1V2_GR3 Filters Recommendation for Device Fabric* section. • Added *P0V8_GR1 Filters Recommendation for Device Fabric* section. • Added *VCCL Filters Recommendation for Device Fabric* section. • Added Table: *Device Guidelines Status for Intel Agilex 7 Devices (M-Series)*. • Added Table: *Intel Agilex 7 AGM (with only F-Tile, R-Tile, or Both) FPGA Package Power Rail Nets and Subsystem Details*. • Added Figure: *Recommended Intel Agilex 7 AGM (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Engineering Sample (ES) Silicon*. • Added Figure: *Example of Board Power Flooding*. • Added Figure: *VCCPLL_NOC LC Filter Recommendation*. • Added Figure: *VCCLPLL_NOC LC Filter Recommendation*. • Added Figure: *VCCPLLDIG_NOC LC Filter Recommendation*. • Updated the *Power Tree* section. • Updated the *Recommended Power Tree for the Agilex 7 Devices with only P-Tile and E-Tile in the Device Packages* section. • Updated the *Recommended Power Tree for the Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and R-Tile) Device Packages*. • Updated the *Rail Merger Requirements* section. • Updated the *Power Nets* section. • Updated the *Agilex 7 Package Power Nets and Subsystems Details* section. • Updated the *Power Rails Tolerance* section. • Updated the *Intel Agilex 7 FPGA Packages Board-Level Decoupling Capacitors Summary* section. • Updated the *P1V8_GR2 (Intel Agilex 7 F-Series and I-Series)/P1V8_GR3 (Intel Agilex 7 M-Series) Filters Recommendation for Device Fabric* section. • Updated the *Agilex 7 FPGA Packages Board-Level Decoupling Capacitors Summary* section. • Updated the *Board Decoupling Capacitors and Power Flooding Guides* section. • Updated the *Board LC Recommended Filters for Noise Reduction in Combined Power Delivery Rails* section. • Updated Table: *Intel Agilex 7 Device Family Transient and Step Load Specifications at Package Pin*. • Updated Table: *Device Guidelines Status for Agilex 7 Devices (I-Series)*. • Updated Table: *Agilex 7 AGF (with only E-Tile and P-Tile), Agilex 7 AGF (with only F-Tile), and Agilex 7 AGI (with only F-Tile, R-Tile, or both) FPGA Package Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex 7 R-Tile Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex 7 F-Tile Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex 7 Device PCB Power Rail Tolerance*. • Updated Table: *Intel Agilex 7 FPGA F-Series, I-Series, and M-Series Decoupling Capacitors Summary*. • Updated Table: *Agilex 7 E-Tile Decoupling Capacitors Summary per Tile*. • Updated Table: *Agilex 7 Device Family Transient and Step Load Specifications at Package Pin*. • Updated Table: *Recommended Inductors and Ferrite Beads for the LC Filter*. • Updated notes (1) and (3) in Figure: *Recommended Agilex 7 AGI or AGF (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon*. • Updated Figure: *VCCADC, VCCPLL_SDM, and VCCPLL_HPS Filter Recommendation*. • Updated Figure: *VCCA_PLL Filter Recommendation*. • Updated Figure: *VCCPLLDIG_SDM Filter Recommendation*. • Updated Figure: *VCCPLLDIG_HPS Filter Recommendation*. • Updated Figure: *Filter Recommendation for VCCCLK_GXP per Tile*. • Updated Figure: *Filter Recommendation for VCCH_GXP per Tile*. • Updated Figure: *Filter Recommendation for VCCRT_GXP per Tile*. • Updated Figure: *Filter Recommendation for VCCRT_GXP per Tile (Low Power Scenario)*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCRT_GXE per Tile*.

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Document Version	Changes
	• Updated Figure: *Connection Requirement and Filter Recommendation for VCCRT_GXE per Tile (Low Power Scenario)*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCRTPLL_GXE per Tile*. • Updated Figure: *Filter Recommendation for VCCH_FGT_GXF per Tile*. • Updated Figure: *Filter Recommendation for VCCCLK_GXF per Tile*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCH_GXR*. • Updated the product family name to "Intel Agilex 7". • Retitled the document from *AN 910: Intel Agilex Power Distribution Network Design Guidelines* to *AN 910: Intel Agilex 7 Power Distribution Network Design Guidelines*. • Removed Table: *Required Rail Connections*. • Removed Figure: *Recommended Intel Agilex 7 AGF014 (with only E-Tile and P-Tile) Power Tree for Engineering Sample (ES) Silicon*. • Removed Figure: *VCCPT LC Filter Recommendation*. • Removed Figure: *VCCPT LC Filter Recommendation (Low Power Scenario)*. • Removed Figure: *Filter Recommendation for VCCERT1_FHT_GXF per Tile*. • Removed Figure: *Filter Recommendation for VCCERT2_FHT_GXF per Tile*. • Removed Figure: *Filter Recommendation for VCCEHT_FHT_GXF per Tile*.
2022.11.30	• Added the *Device Guidelines Status for Intel Agilex Devices* section. • Added Figure: *VCCPT LC Filter Recommendation (Low Power Scenario)*. • Added Figure: *VCCA_PLL LC Filter Recommendation (Low Power Scenario)*. • Added Figure: *Filter Recommendation for VCCH_GXP per Tile (Low Power Scenario)*. • Added Figure: *Filter Recommendation for VCCRT_GXP per Tile (Low Power Scenario)*. • Added Figure: *Connection Requirement and Filter Recommendation for VCCRT_GXE per Tile (Low Power Scenario)*. • Updated the *Board LC Recommended Filters for Noise Reduction in Combined Power Delivery Rails* section. • Updated the *Power Rails Tolerance* section. • Updated the *Intel Agilex FPGA Packages Board-Level Decoupling Capacitors Summary* section. • Updated the *Board Decoupling Capacitors Guide* section. • Updated the *FPGA Core Fabric VCC Voltage Regulator Selection* section. • Updated the *Remote Sense Connections* section. • Updated Table: *Required Rail Connections*. • Updated Table: *Intel Agilex AGF (with only E-Tile and P-Tile), Intel Agilex AGF (with only F-Tile), and Intel Agilex AGI (with only F-Tile, R-Tile, or both) FPGA Package Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex R-Tile Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex Device PCB Power Rail Tolerance*. • Updated Table: *Intel Agilex Device Family Transient and Step Load Specifications at Package Pin*. • Updated Table: *Intel Agilex F-Series and I-Series FPGA Decoupling Capacitors Summary*. • Updated Table: *Intel Agilex E-Tile Decoupling Capacitors Summary per Tile*. • Updated Table: *Intel Agilex P-Tile Decoupling Capacitors Summary per Tile*. • Updated Table: *List of Supported Controllers and Power Stages*. • Updated note (1) in Figure: *Recommended Intel Agilex AGF014 (with only E-Tile and P-Tile) Power Tree for Engineering Sample (ES) Silicon*. • Updated note (1) in Figure: *Recommended Intel Agilex AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices*. • Updated note (2) in Figure: *Recommended Intel Agilex AGI or AGF (with only F-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon*. • Updated Figure: *Filter Recommendation for VCCRT_GXP per Tile*. • Updated Figure: *Filter Recommendation for VCCH_GXP per Tile*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCRTPLL_GXE per Tile*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCH_GXR*.
2022.11.30	Updated Figure: *Recommended Intel Agilex AGI or AGF (with only F-Tile, R-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon*.
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Document Version	Changes
2022.09.22	• Added the *VCCBAT RC Circuit Using the Device Security AES BBRAM Key* section. • Added Figure: *VCCBAT Rail RC Circuit Recommendation*. • Updated note (6) in Figure: *Recommended Intel Agilex AGF014 (with only E-Tile and P-Tile) Power Tree for Engineering Sample (ES) Silicon*. • Updated note (6) in Figure: *Recommended Intel Agilex AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices*. • Updated note (9) in Figure: *Recommended Intel Agilex AGI or AGF (with only F-Tile, R-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon*.
2022.09.06	• Updated Figure: *Recommended Intel Agilex AGF014 (with only E-Tile and P-Tile) Power Tree for Engineering Sample (ES) Silicon*. • Updated Figure: *Recommended Intel Agilex AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices*. • Updated Figure: *Recommended Intel Agilex AGI or AGF (with only F-Tile, R-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon*. • Updated the VCCRCORE system connections description in Table: *Intel Agilex AGF (with only E-Tile and P-Tile), Intel Agilex AGF (with only F-Tile), and Intel Agilex AGI (with only F-Tile, R-Tile, or both) FPGA Package Power Rail Nets and Subsystem Details*.
2022.06.02	• Added Figure: *VCCPLLDIG_SDM Filter Recommendation*. • Added Figure: *VCCPLLDIG_HPS Filter Recommendation*. • Updated the *Power Budget* section. • Updated the *Rail Merger Requirements* section. • Updated the *Intel Agilex R-Tile Board-Level Decoupling Capacitors Summary* section. • Updated the *FPGA Core Fabric VCC Voltage Regulator Selection* section. • Updated the *P-Tile Rail LC Filter Board Scheme and Connection* section. • Updated the *E-Tile Rail LC Filter Board Scheme and Connection* section. • Updated the *F-Tile Rail Board Connection and LC Filter Recommendation under Noisy Voltage Regulator* section. • Updated the *R-Tile Rail Board Connection and LC Filter Recommendation under Noisy Voltage Regulator* section. • Updated the *PCB Voltage Regulator Recommendation for PCB Power Rails* section. • Updated the *VCCA_PLL: Difference between Intel Agilex AGF014 2486A ES and Production Silicons* section. • Updated the *Intel Agilex AGF014/AGF012/AGF008/AGF006, Intel Agilex AGI027/AGI023/AGI022/AGI019, and Intel Agilex AGF027/AGF023/AGF022/AGF019 FPGA Packages Board-Level Decoupling Capacitors Summary* section. • Updated Table: *Required Rail Connections*. • Updated Table: *Intel Agilex AGF (with only E-Tile and P-Tile), Intel Agilex AGF (with only F-Tile), and Intel Agilex AGI (with only F-Tile, R-Tile, or both) FPGA Package Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex Device PCB Power Rail Tolerance*. • Updated Table: *Intel Agilex Device Family Transient and Step Load Specifications at Package Pin*. • Updated Table: *Intel Agilex AGF012/AGF014, Intel Agilex AGI027/AGI023/AGI022, and Intel Agilex AGF027/AGF023/AGF022 FPGA Decoupling Capacitors Summary*. • Updated Table: *Intel Agilex E-Tile Decoupling Capacitors Summary per Tile*. • Updated Table: *Intel Agilex P-Tile Decoupling Capacitors Summary per Tile*. • Updated Table: *List of Supported Controllers and Power Stages*. • Updated Figure: *Recommended Intel Agilex AGF014 (with only E-Tile and P-Tile) Power Tree for Engineering Sample (ES) Silicon*. • Updated Figure: *Recommended Intel Agilex AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices*. • Updated Figure: *Recommended Intel Agilex AGI or AGF (with only F-Tile, R-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon*. • Updated Figure: *VCC/VCCP Core Voltage Regulator Design Implemented on the Intel Agilex AGF014 2486A FPGA Development Kits*. • Updated Figure: *VCCADC, VCCPLL_SDM, and VCCPLL_HPS Filter Recommendation*. • Updated Figure: *VCCPT LC Filter Recommendation*.

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Document Version	Changes
	• Updated Figure: *VCCA_PLL Filter Recommendation*. • Updated Figure: *Filter Recommendation for VCCCLK_GXP per Tile*. • Updated Figure: *Filter Recommendation for VCCH_GXP per Tile*. • Updated Figure: *Filter Recommendation for VCCRT_GXP per Tile*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCRT_GXE per Tile*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCRTPLL_GXE per Tile*. • Updated Figure: *Filter Recommendation for VCCH_FGT_GXF per Tile*. • Updated Figure: *Filter Recommendation for VCCERT_FGT_GXF per Tile*. • Updated Figure: *Filter Recommendation for VCCERT1_FHT_GXF per Tile*. • Updated Figure: *Filter Recommendation for VCCERT2_FHT_GXF per Tile*. • Updated Figure: *Filter Recommendation for VCCEHT_FHT_GXF per Tile*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCH_GXR*.
2022.04.15	• Updated Figure: *Recommended Intel Agilex AGF014 (with only E-Tile and P-Tile) Power Tree for Engineering Sample (ES) Silicon*. • Updated Figure: *Recommended Intel Agilex*. • Updated Figure: *Recommended Intel Agilex AGI or AGF (with only F-Tile, R-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon*. • Updated the VCCBAT board connections and system connections in Table: *Intel Agilex AGF (with only E-Tile and P-Tile), Intel Agilex AGF (with only F-Tile), and Intel Agilex AGI (with only F-Tile, R-Tile, or both) FPGA Package Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex AGF012/AGF014, Intel Agilex AGI027/AGI023/AGI022, and Intel Agilex AGF027/AGF023/AGF022 FPGA Decoupling Capacitors Summary*.
2022.03.08	• Updated the OTN protocol standards for E-tile in the *Intel Agilex E-Tile Board-Level Decoupling Capacitors Summary* section. • Updated the VCCRCORE pin name.
2021.10.18	• Updated Figure: *Recommended Intel Agilex AGF014 (with only E-Tile and P-Tile) Power Tree for Engineering Sample (ES) Silicon*. • Updated Figure: *Recommended Intel Agilex AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices*. • Updated Figure: *Recommended Intel Agilex AGI or AGF (with only F-Tile, R-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon*. • Updated Figure: *VCC/VCCP Core Voltage Regulator Design Implemented on the Intel Agilex AGF014 2486A FPGA Development Kits* to remove typo. • Updated the VCCFUSEWR_SDM voltage in Table: *Intel Agilex Device PCB Power Rail Tolerance*. • Removed the *Power Sequencing Guidelines* chapter and added reference to the *Intel Agilex Power Management User Guide* for the PUS/PDS requirements.
2021.07.23	• Added the *PCB PDN Design Guideline for Unused Tiles* section. • Added Figure: *VCCPT LC Filter Recommendation*. • Updated the *Power Tree* section. • Updated the *Power Tree Updates for Intel Agilex AGF014 2486A Production Silicon* section. • Updated the *Power Nets and Transient Specifications* section. • Updated the *Intel Agilex AGF012/AGF014, Intel Agilex AGI027/AGI023/AGI022, and Intel Agilex AGF027/AGF023/AGF022 FPGA Packages Board-Level Decoupling Capacitors Summary* section. • Updated the *Board Decoupling Capacitors Guide* section. • Updated the *FPGA Core Fabric VCC Voltage Regulator Selection* section. • Updated the *Remote Sense Connections* section. • Updated the *VCC Core Board Current Slew Rate* section. • Updated the *Board LC Recommended Filters for Noise Reduction in Combined Power Delivery Rails* section. • Updated the *P-Tile Rail LC Filter Board Scheme and Connection* section. • Updated the *PCB Voltage Regulator Recommendation for Other Power Rails* section. • Updated the *Board Power Delivery Network Simulations* section. • Updated the *Intel Agilex Device Family PDN Design Summary* section.

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Document Version	Changes
	• Updated Table: *Intel Agilex Device Power Rail Grouping for the PUS Purpose*. • Updated Table: *Required Rail Connections*. • Updated Table: *Intel Agilex AGF (with only E-Tile and P-Tile), Intel Agilex AGF (with only F-Tile), and Intel Agilex AGF Devices (with only E-Tile and P-Tile) Power Tree for AGI (with only F-Tile, R-Tile, or both tiles) FPGA Package Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex E-Tile Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex P-Tile Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex R-Tile Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex F-Tile Power Rail Nets and Subsystem Details*. • Updated Table: *Intel Agilex Device PCB Power Rail Tolerance*. • Updated Table: *Intel Agilex Device Family Transient and Step Load Specifications at Package Pin*. • Updated Table: *Intel Agilex AGF012/AGF014, Intel Agilex AGI027/AGI023/AGI022, and Intel Agilex AGF027/AGF023/AGF022 Decoupling Capacitors Summary*. • Updated Table:*Intel Agilex E-Tile Decoupling Capacitors Summary per Tile*. • Updated Table:*Intel Agilex P-Tile Decoupling Capacitors Summary per Tile*. • Updated Table:*Intel Agilex F-Tile Decoupling Capacitors Summary per Tile*. • Updated Table:*Intel Agilex R-Tile Decoupling Capacitors Summary per Tile*. • Updated Figure: *Recommended Intel Agilex AGF014 (with only E-Tile and P-Tile) Power Tree for Engineering Sample (ES) Silicon*. • Updated Figure: *Recommended Intel Agilex AGF Devices (with only E-Tile and P-Tile) Power Tree for Production Devices*. • Updated Figure: *Recommended Intel Agilex AGI or AGF (with only F-Tile, R-Tile, or both F-Tile and R-Tile) Power Tree for Production Silicon*. • Updated Figure: *Filter Recommendation for VCCH_GXP*. • Updated Figure: *Filter Recommendation for VCCRT_GXP*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCRT_GXE*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCRTPLL_GXE*. • Updated Figure: *Filter Recommendation for VCCH_FGT_GXF*. • Updated Figure: *Connection Requirement and Filter Recommendation for VCCH_GXR*. • Updated Figure: *Time Domain PDN Test Bench Example for Intel Agilex AGF014 VCC Core*. • Edited all "caps" and "decaps" to "capacitors" and "decoupling capacitors".
2021.06.02	• Added VCCH_GXP and VCCCLK_GXP rails to Group 9 Rail 2 column in Table: *Required Rail Connections*. • Updated VCCBAT_SDM power rail name to VCCBAT. • Updated Figure: *Filter Recommendation for VCCRT_GXP*. • Removed VCCBAT from Table: *Intel Agilex Device Power Rail Grouping for the PUS Purpose*. • Removed F-series and I-series references.
2021.03.12	• Added the F-tile and R-tile power rails in the following tables: ☐ Table: *Required Rail Connections* ☐ Table: *Intel Agilex F-Series 2486A and 2581A FPGA Package Power Rail Nets and Subsystem Details* ☐ Table: *Intel Agilex Device Rail Tolerance* ☐ Table: *Intel Agilex Device Family Transient and Step Load Specifications at Package Pin* ☐ Table: *Intel Agilex F-Series 2486A and 2581A FPGA Decoupling Capacitors Summary* • Added Table: *Intel Agilex F-Tile Decoupling Capacitors Summary*. • Added Table: *Intel Agilex R-Tile Decoupling Capacitors Summary*. • Added Table: *Intel Agilex R-Tile Power Rail Nets and Subsystem Details*. • Added Table: *Intel Agilex F-Tile Power Rail Nets and Subsystem Details*. • Added the *Recommended Power Tree for the Intel Agilex I-Series (2957A: R-Tile and F-Tile) Device Packages* section. • Added the *P1V8_GR2 Filters Recommendation for Device Fabric* section. • Added the *F-Tile Rail LC Filter Board Scheme and Connection* section. • Added the *R-Tile Rail LC Filter Board Scheme and Connection* section.

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Document Version	Changes
	• Added Figure: *VCCADC, VCCPLL_SDM, and VCCPLL_HPS Filter Recommendation*. • Added Figure: *VCCA_PLL Filter Recommendation*. • Updated Table: *Intel Agilex Device Power Rail Grouping for the PUS Purpose*. • Updated Figure:*VCC/VCCP Core Voltage Regulator Design Implemented on the Intel Agilex AGF014 2486A FPGA Development Kits*. • Updated Figure: *Filter Recommendation for VCCRT_GXP*. • Updated the *Power-Down Sequencing (PDS)* section. • Updated the *Power Budget* section. • Updated the *Power Rails Tolerance* section. • Updated the *Decoupling Capacitors Recommendation* section. • Updated the *Board Decoupling Capacitors Guide* section. • Updated the *Remote Sense Connections* section. • Updated the section title *FPGA Core Fabric VCCL Voltage Regulator Selection* to *FPGA Core Fabric VCC Voltage Regulator Selection*.
2020.12.18	Updated Figure: *Recommended F-Series 2486A and 2581A Power Tree for Production Silicon* to correct VCCCLX_GXP to VCCCLK_GXP.
2020.12.16	Updated Figure: *Recommended F-Series 2486A and 2581A Power Tree for Production Silicon* to correct the power group for VCCA_PLL from 2 to 3.
2020.12.05	• Updated references to "ES silicon" to "early silicon" throughout the document. • Updated Figure: *Recommended F-Series 2486A Power Tree for Early Silicon*. • Updated Figure: *Recommended F-Series 2486A and 2581A Power Tree for Production Silicon*. • Updated the *Board Decoupling Capacitors Guide* topic: ☐ Updated the VCCH cavity decapacitors on bottom side from 6x 0603 22µ to 4x 0603 22µ. ☐ Added a note to clarify that the power tree and number of decoupling capacitors within cavity on Intel Agilex development kit boards may be slightly different than what has been recommended due to early release of silicon and guideline. • Updated Table: *Intel Agilex F-Series 2486A and 2581A FPGA Decoupling Capacitors Summary*. • Updated Table: *Required Rail Connections*.
2020.08.25	• Updated Figure: *Recommended F-Series 2486A Power Tree for ES Silicon*. • Updated note (1) in Figure: *Recommended F-Series 2486A and 2581A Power Tree for Production Silicon*.
2020.08.19	Initial release.